MS2I2Former: Multiscale Spatial–Spectral Information Interactive Transformer for Hyperspectral Image Classification

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***Abstract*— Transformer models are increasingly used in hyperspectral image (HSI) classification, thanks to their excellent global feature extraction capabilities. However, these networks still need to be improved in recognizing locally complex feature shapes at different scales and handling linear and nonlinear complex correlations between spectral channels. To this end, we propose an innovative multiscale spatial–spectral information interaction transformer (MS2I2Former) architecture. The architecture skillfully integrates lightweight convolution and Transformer, effectively integrates local and global multiscale spatial features and spectral information, and realizes effective interaction between different scales. We design a multiscale spatial–spectral information interaction (MS2I2) module, which efficiently captures multiscale spatial–spectral features by combining deep convolution of convolution kernels of different sizes and orientations with the frequency domain. Based on this, we propose a distance mean cross-covariance representation (DMC2R) based on distance covariance, which aims to deeply explore the linear and nonlinear relationships between different spectral channels. Considering the convolutional kernel parameters and the comprehensive extraction of joint spectral–space features, we developed the hybrid convolution (HC) module, which combines multiple lightweight convolutions to extract deeper spectral-space features. To model complex remote feature relationships, we innovatively propose the multiscale double cross-symmetric transformer (MDCST) module. This module feeds the rich feature representations after multiscale mapping into double cross-symmetric attention (DCSA), which enhances the internal interactions and fusions among features to capture a wider range of feature dependencies. Experimental results show that on four public datasets, MS2I2Former achieves excellent classification results with fewer training samples compared to existing methods. The source code link is available at https://github.com/cslxju/MS2I2Former.**

***Index Terms*— Hyperspectral image (HSI) classification, lightweight convolutions, multiscale spatial–spectral information, transformer.**

I. Introduction

In THE field of remote sensing imagery, there are many advanced technologies, such as super-resolution (SR) technology [1], [2], hyperspectral imaging technology, and multisource data fusion technology, to name a few. However, hyperspectral imaging technology stands as a critical technology within the domain of remote sensing imagery. Compared with ordinary images, hyperspectral images (HSIs) have significant differences. They capture rich land-cover spectral information through tens or even hundreds of consecutive channels and also contain a large amount of spatial information. These properties enable HSI to more accurately characterize objects and effectively distinguish between different land features. As a result, HSIs have been applied in several remote sensing fields, including food quality analysis [3], georeferenced inversion [4], and military reconnaissance [5]. These applications involve numerous tasks in the field of HSI processing, such as image correction, target detection and recognition, spectral reconstruction, land-cover change detection [6], and classification. Among these tasks, classification, as a research hotspot, has the main task of assigning a corresponding category label to each pixel in an image.

In the early studies of HSI classification, most researchers focused their attention mainly on spectral features and adopted manually based traditional methods, such as support vector machines [7], principal component analysis [8], and sparse representation models [9]. However, these methods are not ideal in terms of classification performance due to the neglect of spatial information in HSI and their limited ability to capture detailed image information. Additionally, these methods often require extensive prior knowledge and are time-consuming.

In recent years, the rapid development of deep learning has greatly facilitated HSI classification by automatically learning more representative image features. Currently, a variety of popular deep learning architectures have been successfully applied to HSI classification, including recurrent neural networks [10], convolutional neural networks (CNNs) [11], [12], graph convolutional networks (GCNs) [13], and deep belief networks [14]. In particular, CNNs have become a mainstream and efficient feature extraction method. 1DCNNs capture local and more abstract spectral features by stacking multiple convolutional layers. However, 1DCNNs cannot capture spatial information, so 2DCNNs are used to extract deeper semantic features of HSIs. As network models become more complex, pure CNN-based networks are not ideal in terms of classification performance. To this end, Liang et al. [15] proposed a fast hierarchical 2-D network structure by combining CNN with SimAm. Subsequently, 3DCNN was introduced for HSI

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classification, and Sun et al. [16] proposed a novel lightweight multisource data fusion network by combining 2DCNN and 3DCNN. In contrast, Wang et al. [17] introduced a 3DCNN within the Transformer architecture to improve classification performance by extracting locally detailed features while retaining the Transformer's advantage in handling long-range dependencies.

According to the aforementioned studies, the performance of HSI classification greatly depends on the effective extraction of spatial–spectral features. To this end, a network framework that employs a multiscale [18] two-branch structure for spatial–spectral feature extraction has been validated as an efficient method for HSI classification. To combine the global modeling capability of the Transformer with features at different scales for extracting more discriminative features, a dual-branch multiscale Transformer was proposed in [19] for the HSI classification task. This model utilizes convolution kernels of various scales for feature extraction in HSIs. The multiscale random shape convolutional and adaptive graph convolutional fusion network [20] leverages the strengths of multiscale GCNs and CNNs to obtain spectral and spatial fusion features. To generate complementary spatial–spectral information at the pixel and super-pixel levels, Liu et al. [21] combined CNN and GCN and proposed a hybrid CNN integrated with a multiscale dynamic GCN to extract feature information at multiple scales. Considering the presence of multiscale similar information in HSIs, a multiscale and multiangle approach can effectively model both local and global features. In [22], a multiscale multiangle attention network was constructed to model the relationships between image features at different scales, both locally and globally. As opposed to multiangle networks, Zhang et al. [23] propose a multistage auxiliary learning strategy that learns different features from different stages. Wu et al. [24] propose a multitask multiobjective evolutionary network with joint hyperspectral pansharpening and HSI. In the aforementioned multiscale two-branch structure, it fails to effectively suppress non-critical features while highlighting the important ones. During the extraction of spectral features, the spatial domain is the primary focus for feature extraction, overlooking the potential of transforming the information into other domains for processing. Such a transformation could enhance the recognition and differentiation of spectral features and elevate the significance of global spectral information. Additionally, the current approach to processing spatial features typically involves the use of fixed-size square convolution kernels, which does not adequately concentrate on the features and fails to effectively integrate them through multiscale analysis. As a result, it may also fail to capture both detailed information and the global structure.

In the case of texture similarity and continuous spectral bands, extracting discriminative features presents a challenge. Therefore, recognizing and exploiting similarities between different spectral bands [25] is especially crucial for feature extraction. Cheng et al. [26] combined covariance with the Transformer, which uses unparameterized covariance attention to capture the correlation between different bands, thereby making the network more focused on the useful spectral bands. Fang et al. [27] proposed a new local covariance matrix (CM) method to fully characterize the correlation between different spectral bands. Zhao et al. [28] proposed a CM for HSI feature extraction based on density-peak clustering. This method not only extracts features effectively but also reduces intra-class variation and inter-class interference. In the aforementioned studies on covariance, a common approach is to deal with the CM directly, overlooking the need to analyze the matrix from multiple perspectives. This approach does not effectively simplify complex multidimensional data to highlight the most critical spectral features, thereby simplifying the data processing process. Highlighting the differences between the two covariance matrices and enhancing the model's characterization through nonlinear transformations could help capture subtle spectral variations, which are extremely critical for distinguishing between different feature classes.

The Transformer [29] model has recently become a hot research topic. Initially, it was applied to process sequence-related information, especially in the field of natural language processing. Its key innovation lies in the self-attention mechanism, which can effectively capture global dependencies in sequences, which is one of its significant advantages. Subsequently, the introduction of visual transformer (ViT) brought significant benefits to the field of computer vision. With its excellent tele-dependency processing capability, numerous ViT-based models have been proposed and have demonstrated excellent performance in HSI classification tasks. Shi et al. [30] combined multiscale features with the global modeling capability of Transformer to propose the dual-branch multiscale Transformer. In order to reduce information loss during propagation, Sun et al. [31] proposed the memory enhanced spectral spatial transformer (MASSFormer), leveraging Transformer's powerful processing of spectral sequence information. Considering node embedding and positional encoding as indispensable components in Transformer, Jiang et al. [32] proposed the graph generating structure-aware Transformer to address the issue of the physical meanings of positional encoding being lost in classification tasks. Zhao et al. [33] addressed the problem of parameter overload in processing hyperspectral data and proposed a new approach based on a lightweight ViT, named grouped separable convolutional ViT (GSC-ViT). To fully utilize the information of the spectral dimension, Xu et al. [34] combined 3DCNN and Transformer to propose the dual-branch convolutional Transformer network. Although Transformer-based structures excel in performance, their self-attentive mechanism only learns features from input data within a single feature space. This approach does not fully consider the interaction and fusion between features and is limited by its focus on individual features.

To address the challenges posed by the above networks, we propose MS2I2Former, a multiscale spatial–spectral information interaction transformer (MS2I2Former) network for HSI classification. Specifically, we introduce the multiscale spatial–spectral information interaction (MS2I2) module, which delves into the spatial and spectral details of the image at multiple scales. This module enhances the model's understanding of spectral features by computing and analyzing the distance mean cross-covariance representation (DMC2R), thereby revealing the inter-relationships between different

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spectral channels. Additionally, we have developed a feature processing (FP) module that incorporates a hybrid convolution (HC) block and a multiscale double cross-symmetric transformer (MDCST). The HC block integrates various convolution types, effectively extracting and characterizing the spatial and spectral features of HSI by reducing the number of channels and employing a lightweight convolutional layer, complemented by a channel shuffle operation. The MDCST module first enhances feature representation through multiscale mapping and then models complex feature relationships using a cross-attention mechanism. The main contributions of this work are summarized as follows.

1) We propose a novel multiscale spatial–spectral information interaction Transformer architecture named MS2I2Former, which effectively integrates multiscale spatial and spectral features through lightweight convolutions and Transformer modules, enabling interaction between features of different scales.
2) Aiming at the shortcomings of existing techniques in multiscale spatial–spectral feature extraction, we propose a novel MS2I2 module. The module skillfully integrates convolution kernels of different sizes and orientations for deep convolution, as well as frequency domain analysis, effectively extracting multiscale spatial–spectral features that cover both local details and global structures. Meanwhile, to address the limitations of traditional methods in revealing the correlations between spectral bands (channels), we employ the DMC2R technique to mine the linear and nonlinear correlations among different spectral bands, which provides a deeper understanding of the spectral features for the model.
3) To reduce the number of parameters and computational cost in hyperspectral data processing, we designed a lightweight HC module. This module consists of two branches: the first branch employs a combination of grouped pointwise convolution and grouped convolution, which effectively reduces the number of parameters in the convolution kernel. It emphasizes the local spatial–spectral joint features of the neighboring bands within each group and enhances the exchange of information between channels through channel blending. The second branch integrates 3-D convolution with depthwise convolution (DWConv) to comprehensively extract spatial–spectral information, thereby improving model performance without incurring excessive computational overhead.
4) Aiming at the complexity of feature dependencies in HSI processing and the limitations of existing attention mechanisms in capturing a wide range of dependencies, we propose a novel MDCST module. This module provides a rich feature representation for the subsequent dual cross-symmetric attention mechanism through multiscale mapping, effectively addressing the issue of insufficient feature representation. By employing the dual cross-symmetric attention mechanism, the interaction and information fusion between different features are enhanced, enabling the model to capture a broader spectrum of feature dependencies. Consequently, this improves the model's representation capability and its performance in handling complex scenes.

We have conducted comprehensive experiments on four benchmark HSI classification datasets: Indian Pines, Salinas, Botswana, and WHU-Hi-LongKou. The experimental results demonstrate that our method achieves state-of-the-art performance, outperforming existing HSI classification techniques.

The structure of this article is organized as follows. Section II reviews related work in the field of HSI classification. Section III elaborates on the method proposed by us. Section IV presents the experimental results and provides an in-depth analysis. Finally, Section V summarizes the research work and offers prospects for future research directions.

II. Realted Work

A. CNN-Based HSI Classification Methods

In the field of remote sensing, CNNs have achieved remarkable success in HSI classification tasks due to their excellent local modeling capabilities. Based on this, researchers have proposed numerous CNN-based methods, all of which have demonstrated good performance. Zhai et al. [35] introduced a 1DCNN method designed to extract the rich spectral feature information from HSI data. However, this approach lacks the modeling of spatial information. Liang et al. [15] combined CNN with the SimAm activation function to propose a fast hierarchical 2DCNN network structure for effectively extracting spatial and spectral features. Since HSIs contain more abundant spatial–spectral information, traditional 2DCNNs, due to the limitation of the convolution kernel's receptive field, are still insufficient in feature extraction. In contrast, 3DCNNs possess a 3-D receptive field and capture more complete spatial–spectral information. In order to deeply explore the relationship between spatial and spectral information in HSI, Ben Hamida et al. [36] introduced a novel 3-D deep learning method aimed at synergistic processing of spectral and spatial information. However, as the network depth increases, relying solely on a single convolutional layer is no longer sufficient to achieve satisfactory results. Sun et al. [16] proposed a novel lightweight multisource data fusion network by combining 2DCNN and 3DCNN. Jia et al. [37] introduced a multilayer and multibranch 3-D Gabor CNN, which utilizes multiscale and multidirectional 3-D Gabor filters to effectively characterize the spatial–spectral features within HSI from multiple perspectives. Feng et al. [38] proposed a class-aligned and class-balanced generative domain adaptive method to address the problem that HSI is characterized by class imbalance and complex spatial–spectral distribution. To date, most CNN-based methods have achieved inspiring results. However, the inherent limitation of CNNs in modeling long-range dependencies has hindered further improvements in classification performance.

B. Attention Mechanism-Based HSI Classification Methods

The introduction of attention mechanisms has made a significant contribution to the enhancement of HSI classification performance. As an augmenting mechanism, it strengthens the extraction of useful information while suppressing irrelevant information. Its core objective is to guide the model

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in focusing on information crucial for classification tasks by computing dynamic weights for each part of the global input. Wang et al. [39] proposed a lightweight strategy called attention multilevel feature fusion network aimed at enhancing the resolution of remote sensing images. Fu et al. [40] proposed the attention-enhanced residual and spatial-channel attention network. This method first enhances the residual block through a channel attention mechanism to extract low-level features and then utilizes a spatial-channel attention mechanism to further refine deep features. Xu et al. [41] introduced a cross-attention fusion network that combines CNNs with Transformers. This network is capable of modeling both local and global information within images simultaneously and facilitates interaction between information at different scales through a cross-attention fusion module. In response to the phenomena of "similar spectral signatures for different objects" and "different spectral signatures for the same object" present in HSIs, Zhang et al. [42] proposed the self-attention network. This network effectively integrates spectral and contextual information, allowing the network to focus more on spatial and spectral dimensions through spatial–spectral non-local blocks and multiscale spectral self-attention structures, thus modeling the correlations in both the spatial and spectral domains of HSI. Due to the presence of various shapes and complex land structures in HSI, Xu et al. [43] designed a multiscale and cross-layer attention learning network to fully extract pixel-level local and global multiscale features.

The attention mechanism focuses on important information by assigning weights to each part of the input data, typically relying on additional reference information to guide the weight calculation. Self-attention mechanism, or intra-attention, enables each position in a sequence to simultaneously attend to all other positions, capturing the dependencies between any two positions. Therefore, the self-attention mechanism can capture long-distance dependencies and has strong capabilities for modeling complex internal relationships within sequences. Compared to CNNs, the self-attention mechanism offers greater flexibility, capable of learning dependencies between arbitrary positions without being constrained by a fixed receptive field. In the Transformer model, the self-attention mechanism is a core component; it serves as an independent feature extraction structure and is not used in conjunction with CNNs.

C. Transformer-Based HSI Classification Methods

The emergence of the Transformer has made a significant contribution to the field of computer vision. While CNNs are undoubtedly powerful in terms of local modeling, the Transformer is renowned for its exceptional ability to capture long-distance dependencies. This capability is primarily attributed to its core component—the self-attention mechanism, which excels at handling sequential information. Recently, researchers have designed numerous networks based on the Transformer, achieving commendable results. Ma et al. [44] proposed a spatial pooling Transformer for the problem of HSI classification with noise. The method skillfully fuses the Transformer with an anti-noise algorithm to improve the classification performance. Sun et al. [31] introduced a memory-enhanced network, which effectively fuses spatial–spectral information by converting spatial–spectral features into tokens and inputting these tokens into a memory-enhanced Transformer encoder, while also mitigating the depth decay effect in the model. Zhao et al. [33] designed a lightweight network called GSC-ViT, in which a strategy combining lightweight convolutions with self-attention mechanisms is used to effectively capture global and local spatial–spectral information in HSI. Mohamed et al. [45] proposed a new spectral space Transformer for factorization using a self-supervised pretraining strategy. Shi et al. [46] proposed a multiscale hierarchical Conv-aided Fourier Transformer when studying the token problem in the Transformer. The model skillfully integrates CNN, Transformer, and Fourier transform, and effectively solves the problem of HSI classification.

Although existing methods adopt multiscale strategies, they are still deficient in effectively fusing multiscale features to improve classification performance. Different feature targets may exhibit significant features at different scales, and most of the existing methods only use the same scale for feature extraction, which may lead to problems such as insufficient feature extraction. Transformer's self-attention mechanism mainly captures global information and is weak in extracting local details, which leads to difficulties for the model in extracting local features with complex textures and structures when dealing with HSIs with complex textures and structures. distinguishable local features. Meanwhile, how to balance the relationship between global and local features is still a problem to be solved. In addition, there is a highly nonlinear relationship between spectral channels in HSIs, and Transformer-based methods are not effective in dealing with this nonlinear relationship. The self-attention mechanism, although capable of capturing long-range dependencies, still exhibits limitations in coping with the nonlinear characteristics of the spectral channels.

III. Proposed Method

This section will delve into the MS2I2Former network designed by us. First, we will provide an overview of the overall architecture of the network and its application process in the task of HSI classification. Subsequently, we will dissect the design philosophy of the key components within the MS2I2Former network in detail.

A. MS2I2Former Framework

The overall architecture of the MS2I2Former network is depicted in Fig. 1 and consists of four core components: first, the MS2I2 module; second, two FP modules; third, a global average pooling (GAP) layer; and finally, a SoftMax classifier incorporating a fully connected (FC) layer.

Within the MS2I2Former framework, the network accepts a sequence of small 3-D cubes as input. Owing to the discrepancies across datasets, the spectral (channel) dimensions of each cube vary. In order to enable the multiscale sub-modules within the MS2I2 module to extract spatial and spectral features at various scales more efficiently, we initially apply the spectral–spatial convolution (SSC) block within the

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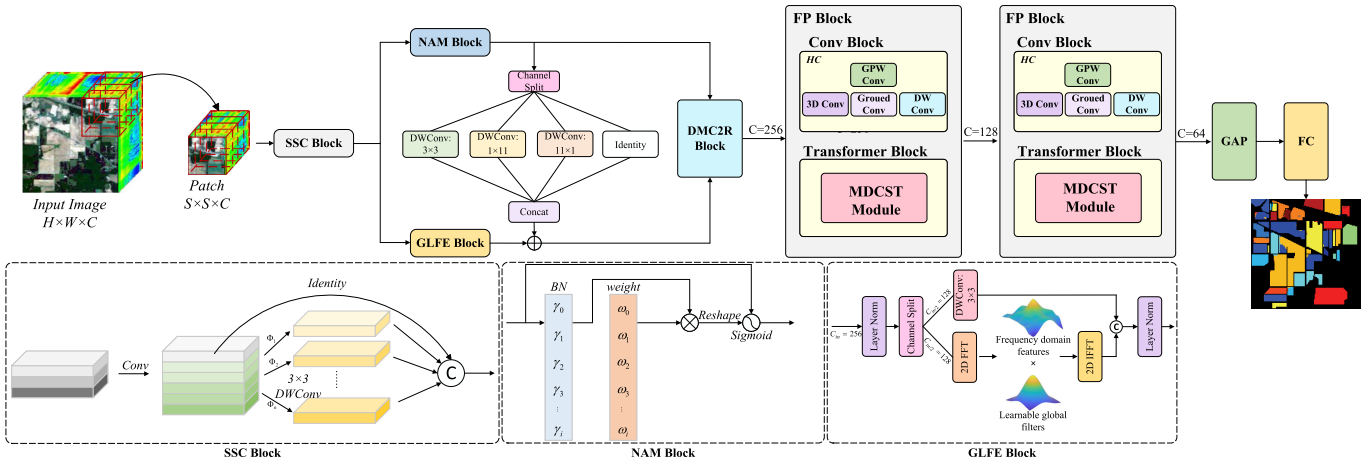


Fig. 1. Proposed MS2I2Former network framework. MS2I2Former mainly consists of the MS2I2 module and two FP modules. The MS2I2 module consists of five sub-modules: the SSC block, the NAM block, the GLFE block, the IDC block, and the DMC2R block. The FP module consists of the HC block and the MDCST block.

MS2I2 module to harmonize the spectral dimensions to a uniform value, and from this, we extract the spatial–spectral features. Following this, we employ the DMC2R approach to analyze the inter-correlation between different spectral bands (channels). Next, the second core component of the model consists of two FP modules: the HC module and the MDCST module. The HC module is composed of four different types of convolutional blocks, including grouped point-wise convolution (GPWConv), group convolution (GroupedConv), 3-D convolution (3DCNN), and DWConv. The primary function of this module is to extract local joint spectral and spatial features and to perform channel-wise interactive shuffling. Following the HC module, the MDCST module is responsible for extracting multiscale local and global spatial features, as well as the interaction and information fusion among different features, through the DCSA. At the GAP layer, the output feature maps are reduced to a 1×1 spatial dimension, then transformed into a 1-D vector, which is fed into a classifier composed of FC layers to obtain the final classification results. The above process is described in Algorithm 1 in a step-by-step manner.

B. MS2I2 Module

In the initial phase, we utilize the SSC block to harmonize the spectral dimensions of the input to a fixed value and extract local spectral–spatial information. Subsequently, by integrating the global–local feature extraction (GLFE) block and the inception DWConv (IDC) block, we extract local and global information across multiple scales. Concurrently, an efficient attention mechanism known as normalization attention mechanism (NAM) is employed to highlight significant features and suppress those that contribute less to the classification task. Finally, to delve deeper into the intrinsic relationships between spectral bands (channels), the extracted features are passed through the DMC2R block. By computing and analyzing the covariance matrices, we uncover the correlations between different spectral channels, thereby providing the model with a deeper understanding of the spectral features.

1) SSC Block: To unify the spectral dimensions of the input and acquire local spectral-spatial information, we have designed an SSC block, as illustrated in Fig. 1. The specific operations are as follows: first, the input image is segmented into small 3-D cubes $X \in \mathbb{R}^{S\times S\times C}$, which are processed through a 1×1 standard convolutional layer to harmonize the spectral dimensions of the input. Subsequently, to avoid the redundancy and high computational cost associated with standard convolutions, we employ 3×3 DWConv in place of standard convolutions to extract local spatial–spectral features. Moreover, to effectively utilize the features from previous layers, we adopt residual connections by channel-wise concatenating the final output with the features from the previous layer. The output of the SSC block is expressed in the following equation:

$$\begin{aligned} Z_{\text{SSC}} &= \text{Concat}[Z_{\text{DWConv}}, Z_{\text{Conv2d}}] \\ Z_{\text{DWConv}} &= \text{DWConv}_{3\times 3}(Z_{\text{Conv2d}}) \\ Z_{\text{Conv2d}} &= \text{Conv2d}_{1\times 1}(X) \end{aligned} \tag{1}$$

where Z_{SSC}, Z_{DWConv}, and Z_{Conv2d} represent the outputs of the SSC block, the DWConv layer, and the Conv layer, respectively. The Concat$(\cdot)$ operation denotes the concatenation of different features along the channel dimension. The number of output channels for both Z_{DWConv} and Z_{Conv2d} is 128, while the number of output channels for Z_{SSC} is 256.

2) NAM Block: Given that local spatial–spectral features are extracted from the SSC block and considering the rich spectral characteristics of HSI data, the features extracted from the spectral dimension (i.e., the channel dimension) may contain some non-significant information. To more effectively suppress these non-significant features, we introduce NAM, as shown in Fig. 1. This module assesses the importance of each spectral band (channel) by analyzing the scaling factors within batch normalization (BN) [47]. The scaling factors reflect the degree of variation within the spectral bands and can be regarded as an indicator of the richness of spectral information for each band. If the scaling factor for a spectral band is large, indicating a high variance, it suggests that the spectral information in that band varies significantly, and thus it is of higher importance. Conversely, spectral bands with smaller scaling factors, indicating less variation, have more

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Algorithm 1 Process of MS2I2Former

Input: The input image is segmented into small 3-D cubes $X \in \mathbb{R}^{S\times S\times C}$, all iterations N.

Output: The classification graph obtained by MS2ISFormer algorithm.

for $i = 1$ *to* N **do**
- Obtain the feature map Z_{SSC} after unifying the spectral dimensions by (1); `// SSC Block`
- Obtain the feature map Z_{NAM} after enhancing or weakening each channel of the input feature map by (2)-(3); `// NAM Block`
- Obtain a richer representation of the features by (4)-(7); to obtain the feature map Z_{GLFE}; `// GLFE Block`
- Obtain the feature map Z_{IDC} containing richer multiscale information and the resultant feature map Z_{Fusion} after fusion of GLFE Block and IDC Block by (8)-(11); `// IDC Block`
- Calculate the two distance covariance matrices A^1 and A^2 by (12)-(13);
- Generate the final distance mean cross-CM A_{DMC2R} by (14)-(17) and A^1 and A^2; `// DMC2R Block`
- Generate the feature map after joint feature extraction by HC blocks GPWConv, GroupedConv, 3DCNN and DWConv by (18)-(20); `// HC Block`
- Generate $Q1, K1, V1$ and $Q2, K2, V2$ after two sets of mapping by (21)-(22); `// Multiscale Mapping`
- Calculate the final output feature map F_{out} by (23)-(25) attention; `// MDCST Block`
- Transform the feature map into 1-D vectors by GAP and feed it into the classifier in the FC layer to obtain the final classification result;
- Calculate the total loss over CrossEntropyLoss;

uniform information and are considered of relatively lower importance. The scaling factors are expressed in the following equation:

$$B_{\text{out}} = \text{BN}(B_{\text{in}}) = \gamma \frac{B_{\text{in}} - \mu_{\text{batch}}}{\sqrt{\sigma^2_{\text{batch}} + \epsilon}} + \beta \tag{2}$$

where B_{in} is the input data, B_{out} is the output data after BN; μ_{batch} and σ^2_{batch} are the mean and variance of the input data within the current batch, respectively; γ and β are the learnable scaling and shifting parameters, respectively; ϵ is a very small constant, which is used to prevent the denominator from being zero.

The NAM block, as illustrated in Fig. 1, involves processing the input feature Z_{SSC} with BN operations. Subsequently, we normalize the absolute values of the weights from the BN layer. Specifically, we first compute the absolute value of each weight, then divide it by the sum of the absolute values of all weights, resulting in a weight coefficient vector. Each element in this vector represents the relative importance of the original weight within the entire set of weights. Following this, we perform an element-wise multiplication of this weight coefficient vector with the Z_{SSC} after BN. This operation enhances or diminishes each channel's feature based on its significance, thereby enhancing the model's capability to learn key features. Next, the adjusted features are compressed using the Sigmoid activation function and multiplied by the result of the residual connection. This approach not only retains a portion of the original input information but also helps prevent the vanishing gradient problem. The expression for implementing the NAM is given by the following equation:

$$Z_{\text{NAM}} = \text{sigmoid}(W_\gamma(\text{BN}(Z_{\text{SSC}}))) \tag{3}$$

where Z_{NAM} represents the output of the NAM block. γ denotes the scaling factor for each spectral band (channel), and the weight value W_γ is calculated as $W_\gamma = \gamma_i / \Sigma_{j=0}\gamma_j$.

3) GLFE Block: To obtain a more rich feature representation, we have designed the GLFE block, as shown in Fig. 1. This module first divides the input feature map with input channels $C_{\text{in}} = 256$ into two parts along the channel dimension, with each part containing $C_{\text{in}/2} = 128$ channels. Subsequently, one part of the channel feature maps is processed through a 3×3 DWConv to handle local features. This approach not only reduces the number of parameters and computational complexity but also effectively captures local spatial and spectral features. The specific operations are expressed in the following equations:

$$Z_{\text{dw1}} = \text{DWConv}_{3\times3}\left(Z^1_{C_{\text{in}/2=128}}\right) \tag{4}$$

$$Z^i_{C_{\text{in}/2=128}} = \text{ChannelSplit}\big(\text{LN}(Z_{\text{SSC}})\big), \quad i = 1, 2 \tag{5}$$

where Z_{dw1}, $Z^i_{C_{\text{in}/2=128}}$ represent the outputs of $\text{DWConv}_{3\times3}(\cdot)$ and ChannelSplit$(\cdot)$, respectively; LN$(\cdot)$ denotes layer normalization.

Second, another portion of the channel's feature maps undergoes a 2-D Fourier transform (2D FFT), converting the local details of the image into the frequency distribution of the entire image, which is a transition from the spatial domain to the frequency domain for processing, in order to capture global information. Specifically, we have set a learnable weight α, which is continuously adjusted during the training process, to modulate the intensity of signals at different frequencies or positions. Subsequently, the feature maps in the frequency domain are multiplied by the learnable weight α, thereby introducing learned global features in the frequency domain and enhancing the model's ability to capture global context. The specific operations are shown in the following equations:

$$Z_{\text{GLFE}} = \text{LN}(\text{Concat}(Z_{\text{dw1}}, Z_{\text{fft2}})) \tag{6}$$

$$Z_{\text{fft2}} = \text{2D IFFT}\left(\text{2D FFT}\left(Z^2_{C_{\text{in}/2=128}}\right) * \alpha\right) \tag{7}$$

where Z_{GLFE} and Z_{fft2} represent the outputs of the GLFE block and after the Fourier transform, respectively; 2D FFT$(\cdot)$ and 2D IFFT$(\cdot)$ denote the Fourier transform and the inverse Fourier transform, respectively; α signifies the learnable weight; Concat$(\cdot)$ indicates concatenation along the channel dimension, restoring the dimension to 256; and LN$(\cdot)$ stands for layer normalization.

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4) IDC Block: To jointly extract and effectively utilize spatial and spectral information and achieve multiscale feature extraction, we introduce the IDC block, as shown in Fig. 1. This block is capable of extracting features at multiple scales by combining different convolutional branches, which aids in identifying geospatial objects of varying sizes. In the task of HSI classification, different geospatial object categories may be similar in spectral features but differ in spatial features, or vice versa. Therefore, the design of the IDC block also considers this feature heterogeneity. By integrating spatial and spectral convolutional branches, it ensures the extraction of rich and diverse features, thereby improving classification accuracy. Specifically, for the input Z_{NAM}, it is divided into four groups along the channel dimension according to a certain ratio, with the specific process shown in the following equation:

$$\begin{aligned} Z_{hw}, Z_w, Z_h, Z_{id} &= \mathrm{ChannelSplit}(Z_{\mathrm{NAM}}) \\ &= Z_{\mathrm{NAM}}^{:,:g}, Z_{\mathrm{NAM}}^{:g:2g}, Z_{\mathrm{NAM}}^{:2g:3g}, Z_{\mathrm{NAM}}^{:3g:} \end{aligned} \tag{8}$$

where g represents the number of channels for each convolutional branch. We determine the number of channels for each branch by $g = r_g C$, where r_g is a set ratio. Subsequently, the separated features are transmitted to different parallel branches, with the specific operation shown in the following equation:

$$\begin{aligned} Z'_{hw} &= \mathrm{DWConv}_{\mathrm{k_m}\times\mathrm{k_m}}{}^{g\to g}(Z_{hw}) \\ Z'_{w} &= \mathrm{DWConv}_{1\times\mathrm{k_v}}{}^{g\to g}(Z_{w}) \\ Z'_{h} &= \mathrm{DWConv}_{\mathrm{k_v}\times 1}{}^{g\to g}(Z_{h}) \\ Z'_{id} &= Z_{id} \end{aligned} \tag{9}$$

where k_m represents the size of the rectangular kernel, which is set to 3 in this case; k_v represents a larger kernel size, which is set to 11 here. In this setup, we ensure that a portion of the original input Z_{id} is directly passed to the next layer, which helps maintain the integrity of the information and may reduce the risk of overfitting. Ultimately, the outputs from each branch are concatenated along the channel dimension to obtain the output Z_{IDC} of the IDC block. The specific result is shown in the following equation:

$$Z_{\mathrm{IDC}} = \mathrm{Concat}\left(Z'_{hw}, Z'_{w}, Z'_{h}, Z'_{id}\right). \tag{10}$$

However, in order to more effectively combine multiscale local details and global context information, construct a more comprehensive and robust feature representation, and thereby improve the classification accuracy and robustness of the model, we propose a fusion strategy. This strategy integrates the GLFE block with the IDC block, not only enhancing the richness and diversity of features but also better integrating the local spatial information and global frequency domain information extracted by the two blocks. This fusion aids the model in understanding image content from different scales and perspectives, compensating for the limitations of single-feature extraction methods. The specific fusion equation is shown as follows:

$$Z_{\mathrm{Fusion}} = Z_{\mathrm{GLFE}} + Z_{\mathrm{IDC}} \tag{11}$$

where $Z_{\mathrm{Fusion}} \in \mathbb{R}^{H\times W\times C}$ denotes the fused output result.

5) DMC2R Block: The preliminary feature extraction phase has already captured a sufficiently rich set of spatial–spectral features. In HSI analysis, characterizing the linear and nonlinear interactions within the spectral domain is a crucial yet intricate task. Leveraging the elegant and effective distance covariance descriptor proposed in [48] and [49], we have developed the DMC2R block, which models the integrated spectral–spatial features and meticulously investigates the relationships among these features, with a particular focus on delineating the linear and nonlinear interactions in the spectral domain.

As illustrated in Figs. 1 and 2, to achieve a more comprehensive spectral-spatial representation, the input to the DMC2R block consists of two distinct sets of sample features, $X^1_{i,j} \in \mathbb{R}^{S\times S\times C}$ and $X^2_{i,j} \in \mathbb{R}^{S\times S\times C}$, which are used as the inputs for constructing the distance CM.

As shown in Fig. 2, for the sake of explanation, we take the first branch as an example to construct the distance CM as proposed in [48] and [49] and further obtain the distance mean cross-CM. The same procedure applies to the second branch. Specifically, the input $X^1_{i,j}$ is first reshaped into a 2-D matrix $X^1 \in \mathbb{R}^{S^2\times C}$. Then, the Euclidean distance between any two bands in X^1 can be expressed as follows:

$$\hat{a}_{pq} = \|x_p - x_q\|, \quad p, q = 1, \ldots, C \tag{12}$$

where $x_p \in \mathbb{R}^{S^2}$ and $x_q \in \mathbb{R}^{S^2}$ represent the pth and qth bands in X^1, respectively, and $\hat{a}_{pq}$ is the Euclidean distance between x_p and x_q. This can be further represented as the Euclidean distance matrix $\hat{A}^1 = (\hat{a}_{pq}) \in \mathbb{S}^+_C$. Ultimately, the distance CM A^1 can be expressed by the following equation:

$$A^1 = \hat{A}^1 - \hat{A}^1_{C\cdot} - \hat{A}^1_{\cdot C} + \hat{A}^1_{\cdot\cdot} \tag{13}$$

where $\hat{A}^1_{(C\cdot)}$, $\hat{A}^1_{(\cdot C)}$, and $\hat{A}^1_{(\cdot\cdot)}$ represent the row-wise mean, column-wise mean, and overall mean of all elements in $\hat{A}^1$, respectively. The same reasoning leads to A^2.

Therefore, we can obtain the respective distance covariance matrices A^1 and A^2 from the two branches. Subsequently, to facilitate the cross-fusion of information from the distance covariance matrices derived from two different sample feature sets, the two distance covariance matrices A^1 and A^2 can be averaged along different directions to obtain four distinct 1-D vectors: A^1_v, A^1_h, A^2_v, and A^2_h. Consequently, the two distance mean cross-covariance matrices can be represented by the following equation:

$$A^1_{mc} = A^1_h \otimes A^2_v \tag{14}$$

$$A^2_{mc} = A^2_h \otimes A^1_v \tag{15}$$

where A^1_{mc} and A^2_{mc} represent the two distance mean cross-covariance matrices, respectively, and the symbol $\otimes$ denotes element-wise multiplication.

Next, in order to reveal the subtle changes in statistical properties of different feature mappings within the HSI, we compute the differences between two mean distance cross-covariance matrices. This step is crucial, as it enables the model to focus on the key spatial relationships necessary for distinguishing between different ground object categories. Subsequently, we employ a hybrid activation

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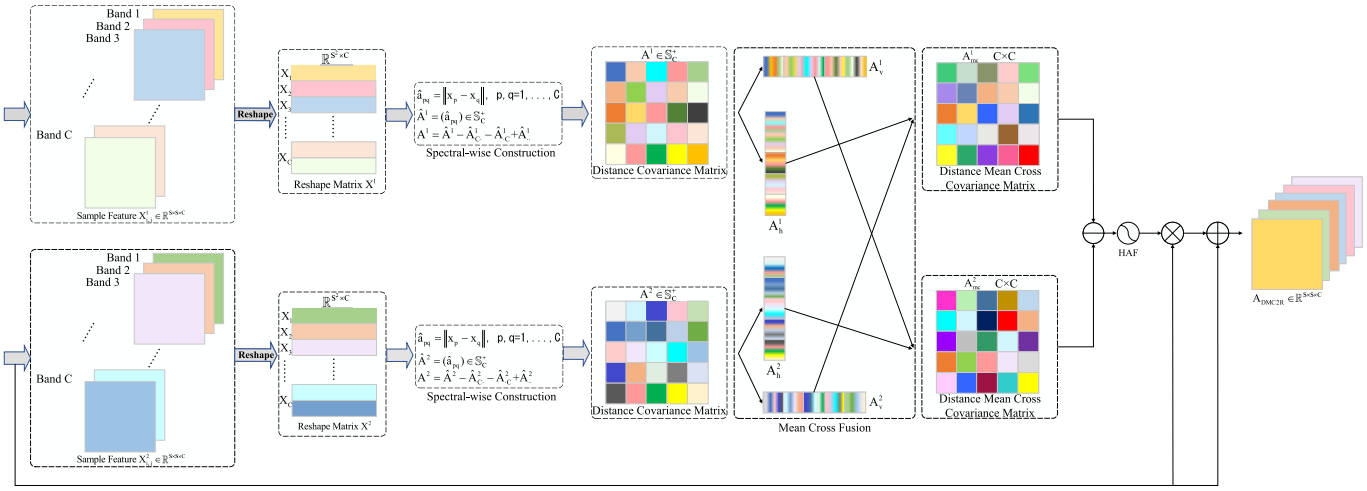


Fig. 2. Structure of DCM2R block. The block receives two different input features and constructs two distance covariance matrices, respectively. Subsequently, two distance mean cross-covariance matrices are generated by the mean cross-fusion technique. After a series of arithmetic processing, the DMC2R matrix is finally obtained.

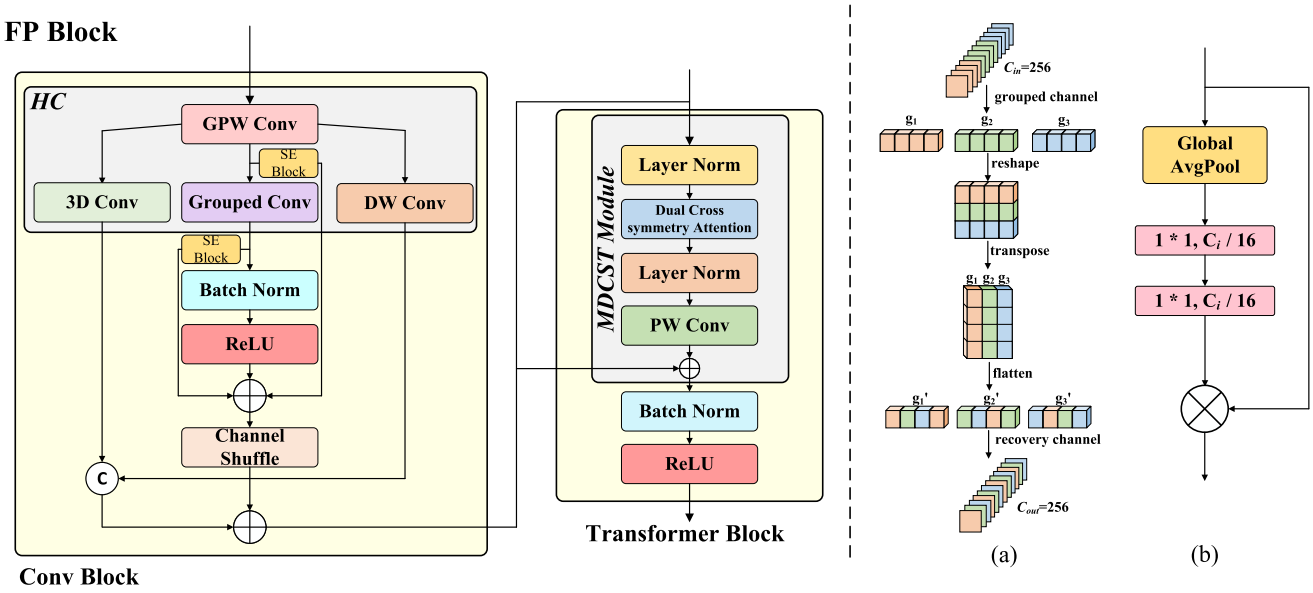


Fig. 3. Structure of FP block. (a) Channel shuffle. (b) SE block.

function (HAF) [26] to emphasize these differences and integrate the differential information with the original features through matrix multiplication operations, thereby enhancing the model's sensitivity to category distinctions. The final feature representation not only contains the original pixel information but also incorporates the additional discriminative power brought by the mean distance cross-covariance differences, providing a more robust feature input for subsequent tasks. The subsequent equations are shown as follows:

$$A_{\mathrm{DMC2R}} = \left(X^2_{i,j} \otimes \left(\mathrm{HAF}\left(A^1_{mc} - A^2_{mc}\right)\right)\right) + X^2_{i,j} \tag{16}$$

$$\mathrm{HAF}(x) = \left(1 - \tanh(x)^2\right) \cdot (\mathrm{sigmoid}(x)(1 - \mathrm{sigmoid}(x)) \tag{17}$$

where A_{DMC2R} represents the final output of the DMC2R block; $X^2_{i,j}$ represents the input of the second branch; and the symbol $\otimes$ denotes element-wise multiplication.

C. Convolutional Block Within the FP Module

Within each FP block, the first to be encountered is the HC block, which primarily consists of four distinct convolutional layers: GPWConv, GroupedConv, 3DCNN, and DWConv. Its architecture is depicted in Fig. 3. First, through a meticulously designed fusion strategy, the combination of GPWConv, GroupedConv, SE block, and channel shuffle operation not only significantly enhances the extraction capability of local spectral–spatial features within groups of adjacent bands but also specifically boosts the saliency of channels that are crucial for classification tasks in the feature maps. Concurrently, the presence of channel shuffle facilitates the exchange of information between different channels, with the random rearrangement of channel order effectively increasing the randomness of training. This aids in manifesting the diversity of sample features, reducing the model's excessive reliance on specific feature representations, and consequently enhancing

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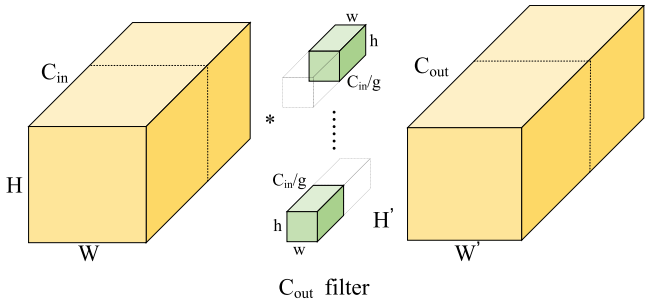


Fig. 4. Structure of GroupedConv.

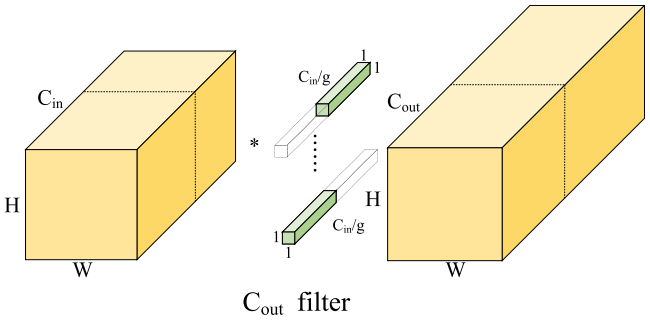


Fig. 5. Structure of grouped pointwise convolution (GPWConv).

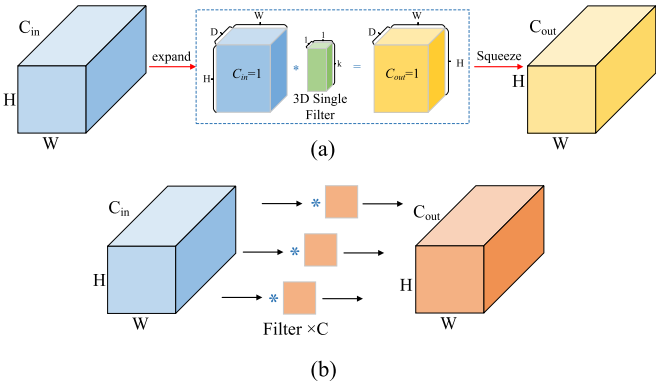


Fig. 6. Structure of 3-D convolution (3DCNN) and depthwise convolution (DWConv). (a) 3DCNN. (b) DWConv.

the model's generalization ability and robustness. Second, the joint application strategy of 3DCNN and DWConv ensures the thorough extraction and representation of both spectral and spatial features.

1) GPWConv and GroupedConv: For GroupedConv, it operates differently from standard convolution. As illustrated in Fig. 4, the input features are evenly divided into g non-overlapping groups along the channel dimension, and then standard convolution is performed on each group. Specifically, if the input features are denoted as $Z \in \mathbb{R}^{C_{\text{in}} \times H \times W}$, Z is split into g groups along the channel dimension. The formula is expressed as follows:

$$Z = \{Z_1, Z_2, \ldots, Z_i, \ldots, Z_g\} \tag{18}$$

where g denotes the number of groups, and the size of the feature map for each group is $Z_i \in \mathbb{R}^{C_{\text{in}} \times H \times W}$. Following this, a standard convolution is applied independently within each group to obtain the output results for each group. These outputs are then concatenated to form the final feature map Z', as indicated in the following equation:

$$Z' = \text{Concat}(Z'_1, Z'_2, \ldots, Z'_i, \ldots, Z'_g). \tag{19}$$

In conventional convolution, the size of the input features is $C_{\text{in}} \times H \times W$, and the size of the output features is $C_{\text{out}} \times H' \times W'$. Therefore, it requires C_{out} convolution kernels of size $C_{\text{in}} \times h \times w$. For grouped convolution, the size of the input and output features remains consistent, but the size of the convolution kernels is reduced to $(C_{\text{in}/g}) \times h \times w$. Consequently, the total number of convolution kernels for conventional convolution is $C_{\text{in}} \times C_{\text{out}} \times h \times w$, while for grouped convolution, the total number of convolution kernels is $C_{\text{in}/g} \times C_{\text{out}} \times h \times w$. In essence, the number of parameters for grouped convolution is $1/g$ of that for conventional convolution.

GPWConv, as a special form of GroupedConv, has a convolution kernel size of 1×1. The main difference from GroupedConv lies in the fact that GPWConv maintains spatial dimension invariance throughout the convolution process. As shown in Fig. 5, compared to the conventional convolution with 1×1 convolution kernels, the number of parameters in GPWConv is reduced to $1/g$ of the latter.

2) 3DCNN and DWConv: For 3DCNN, we combine it with the expand-squeeze strategy, fully leveraging the capability of 3DCNN to automatically learn features, and set the convolution kernel to $1 \times 1 \times 3$ to enhance the extraction of spectral features. Specifically, as shown in Fig. 6(a), during the 3-D convolution, the D dimension of Z is treated as the depth dimension. Therefore, we transform the input features $Z \in \mathbb{R}^{H \times W \times C_{\text{in}}}$ into a 4-D space to obtain $Z_1 \in \mathbb{R}^{H \times W \times D \times 1}$. Subsequently, we also set the number of output channels to 1, and through the convolution operation, we obtain the output $Z_{3D} \in \mathbb{R}^{H \times W \times D \times 1}$. Finally, the squeeze operation is performed to obtain $Z_{3dout} \in \mathbb{R}^{H \times W \times C_{\text{out}}}$, with the specific formula as shown in the following equation:

$$\begin{aligned} Z_1 &= \text{expand}(Z) \in \mathbb{R}^{H \times W \times D \times 1} \\ Z_{3D} &= \text{3DCNN}(Z_1) \in \mathbb{R}^{H \times W \times D \times 1} \\ Z_{3dout} &= \text{squeeze}(Z_{3D}) \in \mathbb{R}^{H \times W \times C_{\text{out}}} \end{aligned} \tag{20}$$

where expand$(\cdot)$ and squeeze$(\cdot)$ denote the expansion and squeezing operations, respectively; 3DCNN$(\cdot)$ denotes the 3-D convolution operation; and $C_{\text{in}} = C_{\text{out}}$.

As shown in Fig. 6(b), during DWConv, each filter acts on only one input channel, which is equivalent to performing g convolution operations independently for each channel. Each convolution operation is one-to-one, i.e., one input channel corresponds to one output channel.

D. MDCST Block Within the FP Module

Although the traditional ViT is able to capture global dependencies, it still has some limitations when dealing with images with distinct local features. To this end, we propose a structure called multiscale double cross-symmetric transformer (MDCST), which consists of two parts: multiscale mapping (MM) and double cross-symmetric attention (DCSA). MM operation utilizes convolutional kernels of different sizes

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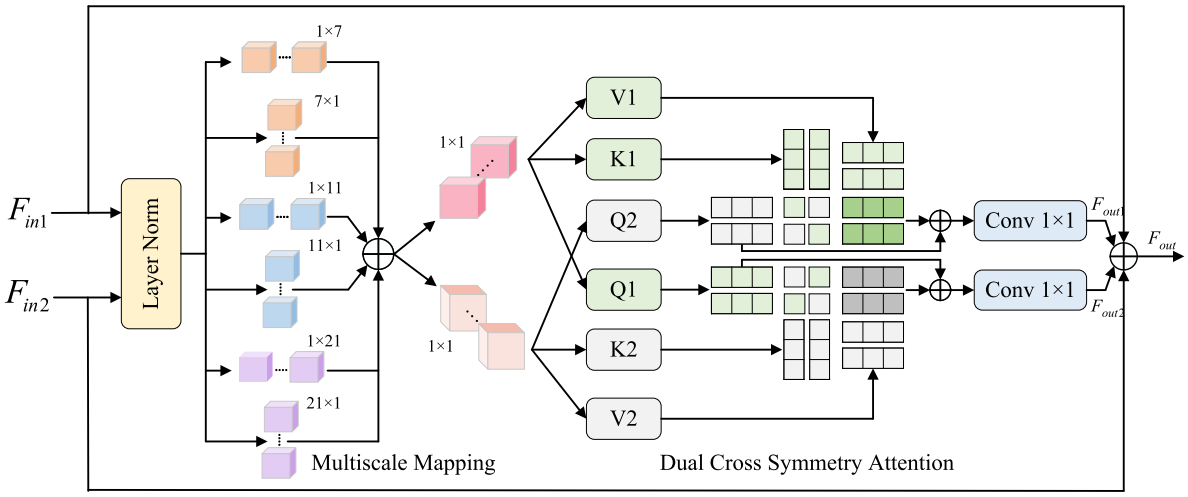


Fig. 7. Structure of MDCST block. The block contains two inputs, each of which is multiscale mapped to generate two different sets of queries (Q), keys (K), and values (V). Subsequently, these two sets of features are processed using the dual cross-symmetric attention mechanism to obtain two outputs. Finally, these two outputs are feature fused with the original two inputs to obtain the final result.

to extract multiscale local features, while DCSA utilizes a special self-attention mechanism to capture global dependencies, thus achieving an effective combination of local features and global information. Thus, the effective combination of local features and global information is realized. The MM operation has been useful in feature screening and compression, alleviating the computational burden of the self-attention mechanism in DCSA and improving the overall computational efficiency. In addition, as shown in Fig. 3, given that HSI relies less on highly abstract semantic information, we replace the multilayer perceptron (MLP) layer with a 1×1 convolutional layer and improve the jump connection mechanism in the Transformer block. These adjustments help remove unnecessary nonlinear transformations and further reduce the overall number of parameters of the model by performing feature learning only through a single Transformer block.

As shown in Fig. 7, we first map the two input features F_{in1}, F_{in2} to a uniform scale by multiscale mapping. Specifically, we design vertical bar convolutions with different scales to capture the multiscale feature information, and then we fuse these different scales weightedly to generate a rich feature representation. Then, we use 1×1 convolution to map them to the uniform scale matrix Q, K, and V for the subsequent attention mechanism.

The two input features F_{in1}, F_{in2} yield two sets of matrices (Q_1, K_1, V_1) and (Q_2, K_2, V_2), respectively. Subsequently, we propose a double cross-symmetric attention (DCSA) approach, which involves two different branches querying each other and their own key-value pairs in a cross-symmetric manner to compute the attention, which enhances the interaction and fusion of information between the different features and then performs the weighting of the features to ultimately achieve the feature selection. Finally, the two output features F_{out1}, F_{out2} are summed and fused with the original input features F_{in1}, F_{in2} to achieve residual connectivity, which effectively fuses different levels of feature information and enhances the model's learning ability and generalization performance.

Through DCSA, it is a special attentional mechanism that allows the model to dynamically adjust the attentional weights based on the feature attributes of HSIs. In this way, the model is able to recognize and enhance features that are relevant to the task, while suppressing or ignoring features that are not relevant to the task. This interactive crossover mechanism helps the model to pay more attention to those features that are critical to the classification task during the feature selection process, thus improving the model's classification performance.

The exact derivation is as follows: if the input $X \in \mathbb{R}^{C\times H\times W}$, we define the multiscale strip vertical convolution operation as shown in the following equation:

$$\begin{aligned} Q, K, V &= \mathrm{F_{MM}}(X) \\ &= \mathrm{Conv2d}_{1\times 1}\big(\mathrm{SVC}_7\big(\mathrm{LN}(X)\big) \\ &\quad + \mathrm{SVC}_{11}\big(\mathrm{LN}(X)\big) \\ &\quad + \mathrm{SVC}_{21}\big(\mathrm{LN}(X)\big)\big) \end{aligned} \tag{21}$$

where $\mathrm{SVC_k}(\cdot)$ denotes the bar convolution of size $1 \times k$ and $k \times 1$; $\mathrm{LN}(\cdot)$ denotes layer normalization; $\mathrm{F_{MM}}(\cdot)$ denotes the multiscale mapping operation; and $Q, K, V \in \mathbb{R}^{C\times H\times W}$ is the matrix obtained after multiscale mapping.

The operation of the entire multiscale mapping for the strip vertical convolution is depicted in the following equation:

$$\begin{cases} Q_1, K_1, V_1 = \mathrm{F_{MM}}(F_{\text{in1}}) \\ Q_2, K_2, V_2 = \mathrm{F_{MM}}(F_{\text{in2}}). \end{cases} \tag{22}$$

The attention equation is shown as follows:

$$\mathrm{Attn}(Q, K, V) = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{23}$$

where $d = C \times H \times W$. The two outputs corresponding to F_{in1} and F_{in2} are calculated as follows:

$$\begin{cases} F_{\text{out1}} = \mathrm{Conv2d}_{1\times 1}(\mathrm{Attn}(Q_2, K_1, V_1) + Q_2) \\ F_{\text{out2}} = \mathrm{Conv2d}_{1\times 1}(\mathrm{Attn}(Q_1, K_2, V_2) + Q_1). \end{cases} \tag{24}$$

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TABLE I

LAND-COVER CATEGORIES OF THE INDIAN PINES DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Valid	Test
C01	Alfalfa	3	2	41
C02	Corn-notill	71	72	1285
C03	Corn-mintill	42	41	747
C04	Corn	12	12	213
C05	Grass-pasture	24	24	435
C06	Grass-trees	37	36	657
C07	Grass-pasture-mowed	2	1	25
C08	Hay-windrowed	24	24	430
C09	Oats	1	1	18
C10	Soybean-notill	49	48	875
C11	Soybean-mintill	122	123	2210
C12	Soybean-clean	30	29	534
C13	Wheat	10	10	185
C14	Woods	63	63	1139
C15	Buildings-Grass-Trees-Drives	19	20	347
C16	Stone-Steel-Towers	4	5	84
	Total	513	511	9225

The output operation of DCSA is shown in the following equation:

$$F_{\text{out}} = F_{\text{in1}} + F_{\text{in2}} + F_{\text{out1}} + F_{\text{out2}} \tag{25}$$

where F_{out1} and F_{out2} are the outputs corresponding to the original input features F_{in1} and F_{in2}; $F_{\text{out}} \in \mathbb{R}^{C\times H\times W}$ is the output tensor of the DCSA.

IV. EXPERIMENTS

In this section, in order to evaluate the effectiveness of the proposed MS2I2Former framework, we conduct comprehensive experiments on four publicly available datasets and compare them with state-of-the-art approaches. First, the four datasets used are described in detail. Then, the configuration of the experiments is described and analyzed. Next, the classification results and the analysis process of our method versus the compared methods are shown. Finally, a detailed ablation experiment was performed to validate the effectiveness of each module.

A. Four Publicly Available Datasets

1) Indian Pines: The Indian Pines dataset was acquired by the AVIRIS instrument of the National Aeronautics and Space Administration (NASA) in 1992 over the northwestern part of Indiana. This dataset comprises an image of 145 × 145 pixels, encompassing 16 distinct land-cover categories, with a spectral range from 0.4 to 2.5 μm, consisting of 220 contiguous spectral bands. Typically, 20 noise bands are removed, leaving 200 bands for research purposes. As illustrated in Fig. 8, the false-color composite image and the ground-truth image of land-cover categories for the Indian Pines dataset are presented. The dataset contains 10 249 labeled samples, which are distributed as follows: 5% for the training set, 5% for the validation set, and 90% for the test set, with the specific allocation detailed in Table I.

2) Salinas: The Salinas dataset was captured by the AVIRIS imaging spectrometer over the Salinas region in California, United States. It possesses a spatial resolution of 3.7 m, and the dataset dimensions are 512 × 217 pixels. Similar to the Indian Pines dataset, 20 interference bands are removed, resulting in a total of 204 bands being retained. Fig. 9 displays the false-color composite image and the corresponding ground-truth image of the Salinas dataset. This dataset encompasses 54 129 labeled samples across 16 different crop categories, with the training samples constituting 1%, the validation samples 1%, and the test samples 98% of the total, as detailed in Table II.

3) Botswanan: The Botswana dataset was acquired by the Earth Observing-1 (EO-1) satellite of the National Aeronautics and Space Administration (NASA) in May 2001, covering the Okavango Delta region of Botswana. The dataset dimensions are 1476 × 256 pixels, with a wavelength range spanning from 0.4 to 2.5 μm and a spatial resolution of approximately 20 m. Fig. 10 presents the false-color composite image and the corresponding ground-truth image of the Botswana dataset. Out of the 242 spectral bands, the following five noise bands were excluded: (1–9, 56–81, 98–101, 120–133, 165–186),

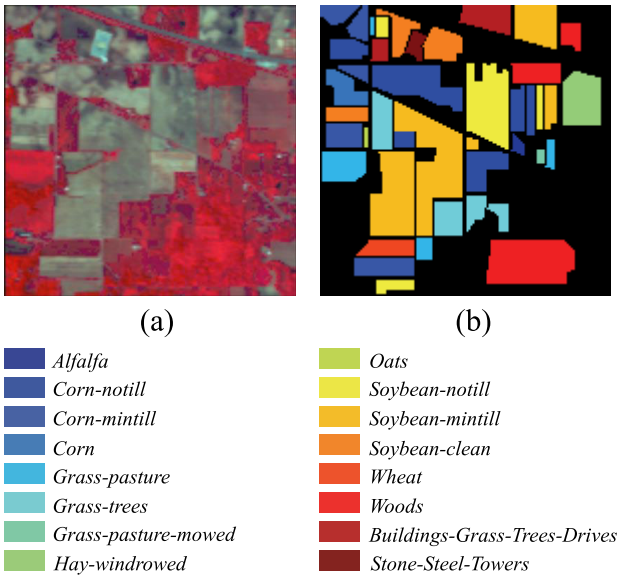


Fig. 8. (a) False-color composition of the AVIRIS Indian Pines scene. (b) Ground-truth map containing 16 mutually exclusive land-cover classes.

TABLE II

LAND-COVER CATEGORIES OF THE SALINAS DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Valid	Test
C01	Brocoli-green-weeds-1	20	20	1969
C02	Brocoli-green-weeds-2	37	37	6352
C03	Fallow	20	20	1936
C04	Fallow-rough-plow	14	14	1366
C05	Fallow-smooth	27	27	2624
C06	Stubble	40	39	3880
C07	Celery	36	36	3507
C08	Grapes-untrained	113	112	11046
C09	Soil-vinyard-develop	62	62	6079
C10	Corn-senesced-green-weeds	33	33	3212
C11	Lettuce-romaine-4wk	11	10	1047
C12	Lettuce-romaine-5wk	20	19	1888
C13	Lettuce-romaine-6wk	9	9	898
C14	Lettuce-romaine-7wk	10	11	1049
C15	Vinyard-untrained	73	72	7123
C16	Vinyard-vertical-trellis	18	18	1771
	Total	543	539	53047

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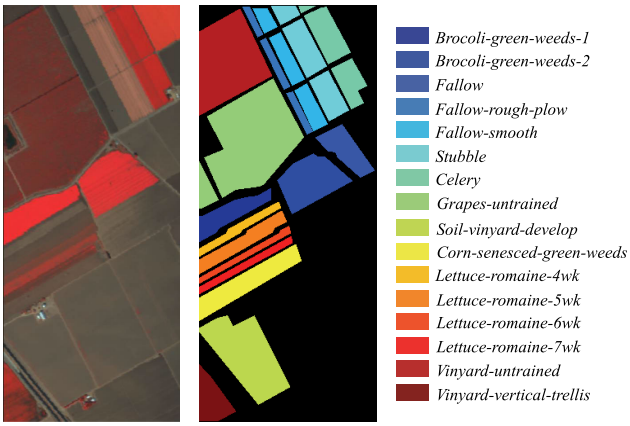


Fig. 9. (a) False-color composition of the AVIRIS Salinas scene. (b) Ground-truth map containing 16 mutually exclusive land-cover classes.

TABLE III

LAND-COVER CATEGORIES OF THE BOTSWANA DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Valid	Test
C01	Water	14	13	243
C02	Hippo-grass	5	5	91
C03	Hippo grass-1	13	12	226
C04	Hippo grass-2	11	10	194
C05	Reeds	14	13	242
C06	Riparian	14	13	242
C07	Firescar	13	13	233
C08	Island interior	10	10	183
C09	Acacia woodlands	16	15	283
C10	Acacia shrulands	13	12	223
C11	Acacia grasslands	15	15	275
C12	Short mopanc	9	9	163
C13	Mixed mopanc	14	13	241
C14	Chalcedony	5	5	85
	Total	166	158	2924

resulting in a retention of 145 bands. The study area comprises 14 different land-cover types, including seasonal wetlands, intermittent wetlands, and the dry woodlands at the distal end of the delta. The total number of samples is 3248. In the experimental setup, the dataset was divided into training (5%), validation (5%), and test (90%) sets, with the specific allocation detailed in Table III.

4) WHU-Hi-LongKou: The WHU-Hi-LongKou dataset was collected over a town in Hubei, China, utilizing a hyperspectral imaging sensor mounted on an unmanned aerial vehicle (UAV). This dataset primarily encompasses six major crop categories: corn, cotton, sesame, broadleaf soybean, narrowleaf soybean, and rice. Fig. 11 displays the false-color composite image and the corresponding ground-truth image of the WHU-Hi-LongKou dataset. The acquired image has dimensions of 550 × 400 pixels, with a spectral range from 0.4 to 1.0 μm, comprising 270 spectral bands, and a spatial resolution of approximately 0.463 m. The total number of samples is 204 542. In the experimental protocol, the dataset was partitioned into training (0.2%), validation (0.2%), and test (99.6%) sets, with the specific distribution detailed in Table IV.

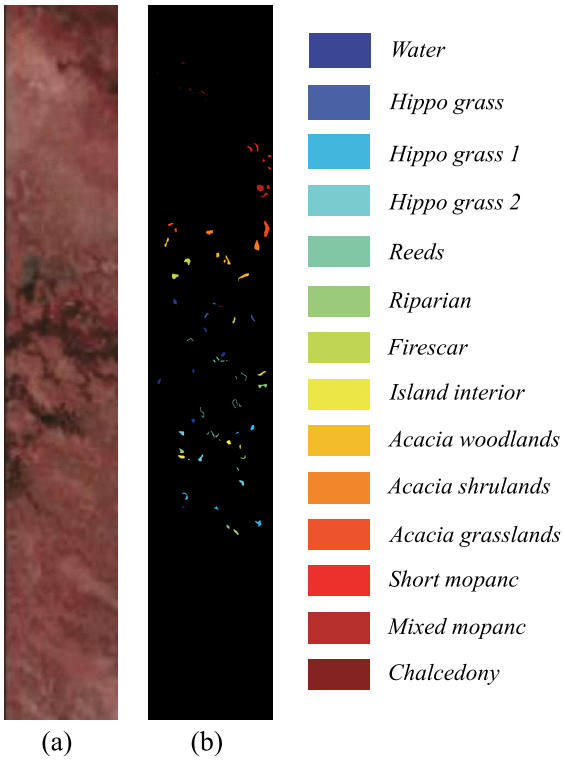


Fig. 10. (a) False-color composition of the AVIRIS Botswana scene. (b) Ground-truth map containing 14 mutually exclusive land-cover classes.

TABLE IV

LAND-COVER CATEGORIES OF THE WHU-HI-LONGKOU DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Valid	Test
C1	Corn	69	69	34373
C2	Cotton	17	16	8341
C3	Sesame	6	6	3019
C4	Broad-leaf soybean	127	126	62959
C5	Narrow-leaf soybean	9	8	4134
C6	Rice	24	23	11807
C7	Water	134	134	66788
C8	Roads and houses	15	14	7095
C9	Mixed weed	11	10	5208
	Total	412	406	203724

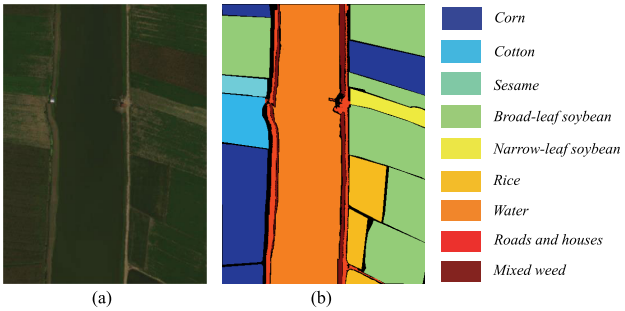


Fig. 11. (a) False-color composition of the AVIRIS WHU-Hi-LongKou scene. (b) Ground-truth map containing nine mutually exclusive land-cover classes.

B. Experimental Setup

1) Configuration: Our proposed MS2I2Former was developed based on the PyTorch framework using Python version 3.7.13 and PyTorch version 1.11. All experiments were done

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on a server equipped with GeForce RTX 3090 GPUs. For the MS2I2Former method, we used the AdamW optimizer to update the network parameters, the loss was the cross-entropy loss, the learning rate was set to 0.0005, the weight decay was 0.05, the batch size was 128, and the size of each patch was 8. In order to reduce the effect of randomness in the experiments and to ensure the stability of the results, we repeated the experiments ten times each time and calculated the average of the results. Due to the varying characteristics of different datasets, the feature extraction ability of the network will also vary.

2) Evaluation Indicators: To reinforce the credibility of our method's performance, we have utilized three comprehensive evaluation indicators: overall accuracy (OA), average accuracy (AA), and Kappa (k). The higher the values of these indicators, the more superior the classification performance of the model.

3) Comparison Methods: In order to highlight the state-of-the-art of MS2I2Former, we compare it with leading methods based on CNN and Transformer architectures, as shown in the following details.

1) *2DCNN [50]:* This network model is mainly composed of 2-D convolution and FC layers. Each convolution is followed by a maximum pooling layer and finally the output is obtained through the FC layer.
2) *3DCNN [36]:* This network structure has a 3-D convolutional core, supplemented by 3-D filler layers and residual blocks, with FC layers and linear layers for classification connected at the end.
3) *SSFTT [51]:* This network structure contains 3-D and 2-D convolutional layers for extracting the underlying features, a Gaussian feature-weighted labeling layer for implementing the feature transformation, a Transformer encoder layer, and a linear layer for classification.
4) *GAHT [52]:* The network is divided into three stages, each of which consists of a 1×1 convolution embedded in grouped pixels and a Transformer encoder module, and the stages are followed by a GAP layer and a linear layer for classification.
5) *MorphFormer [53]:* This network structure fuses 3-D convolution with 2-D convolution, introduces an innovative morphological attention mechanism, and has a linear layer for classification as the output.
6) *SpectralFormer [54]:* This network structure contains a grouped spectral embedding layer for capturing features from neighboring spectral bands, an innovative Transformer encoder fusion layer, and a linear layer for classification.
7) *FactoFormer [45]:* The network has a two-branch design: the main branches are a spectral Transformer layer and a spatial Transformer layer. The spectral Transformer layer has a built-in spectral attention mechanism, while the spatial Transformer layer has a built-in spatial attention mechanism, and the network ends with a linear layer for classification.
8) *MHCFormer [46]:* The network structure consists of multiple 3-D convolutional layers, a 2-D heterogeneous convolutional layer, a spectral spatial attention layer, three Transformer encoder layers, a GAP layer, and a linear layer for classification.

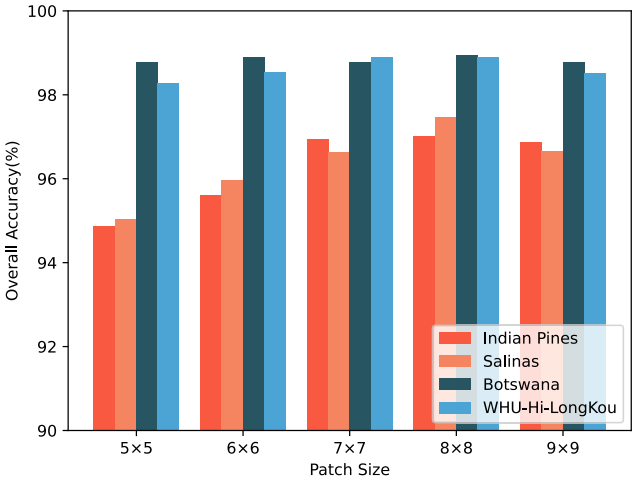


Fig. 12. Impact of five different patch sizes on network performance on four datasets.

9) *SPTNet [44]:* The core of the network consists of 3-D and 2-D convolutional layers for spectral feature extraction, followed by three spatially pooled Transformer layers, followed by a GAP layer, and an FC layer for classification.

C. Parameter Analysis

In order to gain a deeper understanding of the specific impact of different parameter configurations on the performance of classification models, we meticulously evaluated the modulation effects of multiple patch size and learning rate settings on the classification effectiveness of the MS2I2Former model on four different datasets. Through this series of experiments, we aim to reveal which parameter combinations optimally improve the classification accuracy of the model.

1) Patch Size: To investigate the impact of patch size on classification performance, we tested five different patch sizes—5, 6, 7, 8, and 9—while keeping all other parameters constant. As illustrated in Fig. 12, the OA performance of these five patch sizes across four datasets is presented. It is observed that on the Indian Pines, Salinas, and WHU-Hi-LongKou datasets, the OA values exhibit a trend of first increasing and then decreasing with the increase in patch size, with the optimal performance achieved at a patch size of 8. On the Botswana dataset, the OA values show an alternating pattern of increase and decrease, also reaching the highest value at a patch size of 8. Based on these results, we uniformly set the patch size to 8 for all four datasets.

2) Learning Rate: In deep learning, choosing the right learning rate is crucial for network models. A learning rate that is too large or too small can prevent the model from reaching the optimal state, which in turn affects the convergence speed of the model. As shown in Fig. 13, we examine six different learning rates: 0.01, 0.005, 0.0001, 0.0005, 0.00001, and 0.00005. It is obvious from the figure that the performance on the four datasets is mutually superior when the learning rates are 0.0001 and 0.0005. In particular, on the Indian Pines dataset, the performance gap is significant when the learning rate of 0.0001 is compared to the learning rate of 0.0005. After comprehensive evaluation, we decided to set the learning rate at 0.0005.

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TABLE V

PERFORMANCE COMPARISON OF TEN METHODS ON INDIAN PINES DATASET

Class No.	CNNs		Transformers									Our Methods
	2DCNN [50]	3DCNN [36]	SSFTT [51]	GAHT [52]	MorphFormer [53]	SpectralFormer [54]	FactoFormer [45]	MHCFormer [46]	SPTNet [44]	DSNet [55]	MASSFormer [31]	Ours
1	67.17±25.42	55.00±18.34	63.91±20.40	39.57±20.62	70.22±18.68	16.74±12.83	50.87±13.55	**78.48±22.07**	68.91±19.88	42.61±16.36	66.09±17.48	74.35±22.06
2	86.07±2.10	80.52±3.90	89.67±2.41	93.26±2.84	94.09±1.98	68.52±3.03	89.53±1.66	**96.84±1.16**	92.99±1.94	94.66±0.57	92.77±1.91	96.50±1.09
3	88.95±5.94	80.11±5.80	90.39±4.10	93.51±2.27	95.23±3.11	57.11±7.37	94.24±1.82	96.75±1.42	95.59±1.94	96.19±0.90	95.95±1.69	**97.82±0.96**
4	89.83±5.93	74.51±4.97	90.30±2.66	87.93±5.35	93.38±6.39	53.54±9.75	87.38±5.67	**95.36±3.58**	94.64±6.07	88.27±5.28	87.22±6.70	92.66±5.70
5	91.33±4.36	87.02±3.38	92.42±3.37	92.30±3.35	92.67±3.06	57.78±14.44	87.12±5.04	**96.05±2.31**	92.32±3.45	92.88±3.48	91.37±3.44	94.04±3.44
6	96.85±2.01	97.41±1.72	97.86±1.57	99.36±0.59	98.38±1.22	91.47±4.62	97.03±1.86	**99.55±0.36**	98.93±1.31	98.77±0.59	97.59±1.94	98.96±0.77
7	87.14±29.37	73.21±25.87	88.93±18.66	60.00±36.13	87.86±14.21	40.00±33.00	74.29±23.24	69.29±25.71	85.36±11.57	86.43±7.95	98.93±2.29	**96.43±5.05**
8	96.57±3.89	98.56±1.14	99.54±0.79	99.98±0.06	99.60±0.68	96.46±3.34	99.04±0.97	**100.00±0.00**	99.48±0.78	99.29±0.93	99.75±0.33	99.87±0.27
9	45.50±27.61	42.50±6.80	51.00±21.31	17.00±12.88	50.50±18.90	31.50±20.50	54.50±11.06	38.50±22.14	66.50±14.67	57.00±10.77	**82.00±16.76**	80.50±15.88
10	84.96±5.48	76.45±6.42	88.17±4.98	90.08±3.45	93.48±3.03	71.93±4.03	88.12±3.57	92.98±33.78	90.02±5.37	92.70±1.14	93.18±2.46	**94.88±2.07**
11	92.99±2.88	88.81±4.09	95.60±1.43	95.26±2.05	97.18±0.88	83.18±1.73	93.76±2.24	97.82±0.86	95.88±1.27	**98.22±0.54**	96.75±1.39	98.02±0.55
12	81.21±5.84	75.72±4.02	86.90±4.61	90.64±2.63	90.22±3.17	41.96±5.16	82.06±5.64	95.35±2.12	90.35±5.01	94.00±1.43	86.66±5.85	**94.87±1.66**
13	99.27±0.79	99.07±0.99	99.66±0.44	98.98±1.12	99.41±0.75	98.24±1.43	98.24±2.80	99.32±0.79	99.46±0.63	99.61±0.37	98.05±2.36	**99.85±0.22**
14	96.36±1.39	96.74±1.21	95.83±2.26	98.09±0.98	98.89±0.87	93.00±4.26	96.02±2.62	**99.49±0.47**	97.79±1.81	98.31±0.55	97.64±1.53	98.54±0.81
15	87.67±10.11	83.96±4.21	85.52±5.44	90.91±4.68	94.38±2.69	70.88±5.19	78.99±7.31	93.16±4.38	87.59±5.40	90.21±5.25	91.71±6.53	**96.53±1.56**
16	**97.31±2.77**	90.00±10.69	93.98±7.95	94.41±5.58	90.75±7.76	96.24±5.32	91.94±10.17	87.10±10.11	97.20±3.01	96.56±6.36	91.94±6.18	94.62±6.97
OA(%)	90.70±2.41	86.40±1.19	92.68±0.53	94.04±0.63	95.59±0.59	75.55±2.19	91.49±0.72	96.76±0.48	94.57±1.05	95.76±0.36	94.67±0.95	**97.02±0.47**
AA(%)	86.82±5.58	81.23±2.77	88.10±2.76	83.83±2.80	90.39±1.22	66.78±3.70	85.19±2.04	89.75±2.72	90.81±1.46	89.11±1.52	91.72±1.98	**94.28±1.86**
$k \times 100$	89.39±2.75	84.44±1.36	91.65±0.61	93.19±0.72	94.97±0.68	72.02±2.55	90.29±0.82	96.31±0.55	93.80±1.20	95.16±0.41	93.93±1.08	**96.60±0.54**

TABLE VI

PERFORMANCE COMPARISON OF TEN METHODS ON SALINAS DATASET

Class No.	CNNs		Transformers									Our Methods
	2DCNN [50]	3DCNN [36]	SSFTT [51]	GAHT [52]	MorphFormer [53]	SpectralFormer [54]	FactoFormer [45]	MHCFormer [46]	SPTNet [44]	DSNet [55]	MASSFormer [31]	Ours
1	55.88±45.18	98.29±1.28	99.24±1.64	99.80±0.35	99.41±1.19	94.34±1.02	98.89±1.79	98.85±2.23	98.63±1.68	**99.92±0.16**	95.94±4.31	99.74±0.74
2	91.95±13.31	99.76±0.26	99.72±0.45	99.89±0.15	99.87±0.17	99.10±0.63	98.63±1.01	99.76±0.39	98.90±1.81	99.67±0.30	99.47±0.78	**99.99±0.01**
3	68.00±26.72	89.28±4.34	95.94±4.49	98.41±2.13	96.96±2.74	89.08±4.81	95.39±1.92	96.99±2.27	97.48±1.31	95.63±6.00	95.62±1.97	**99.59±0.44**
4	88.65±29.62	98.87±0.94	98.51±1.85	99.33±0.61	99.04±1.73	96.18±1.49	95.93±4.23	99.32±1.35	98.68±1.72	99.28±0.44	98.61±0.66	**99.76±0.28**
5	90.10±9.84	93.75±2.45	94.00±5.10	98.72±1.07	96.37±2.17	86.90±6.04	95.15±3.49	96.74±3.69	95.10±4.27	97.39±0.33	97.09±2.24	**98.83±0.87**
6	99.51±0.71	99.93±0.18	99.98±0.02	99.94±0.13	**99.99±0.02**	99.28±0.80	99.74±0.73	99.93±0.22	**99.99±0.02**	99.93±0.11	99.50±0.73	99.96±0.08
7	96.93±5.24	99.67±0.21	99.77±0.22	99.70±0.25	99.40±0.54	98.31±0.60	96.46±2.56	99.61±0.82	99.73±0.21	99.70±0.10	99.62±0.30	**99.97±0.04**
8	88.13±5.26	85.20±1.47	88.32±2.14	91.36±2.81	94.21±1.37	83.62±2.87	87.93±2.99	92.32±2.68	91.91±1.35	90.39±5.32	92.34±2.56	**94.45±2.25**
9	97.02±2.92	98.84±1.06	99.54±0.46	99.58±0.26	99.53±0.68	96.73±1.64	98.88±1.65	99.81±0.23	99.40±0.56	99.16±1.17	99.35±0.43	**99.84±0.40**
10	88.40±16.96	90.10±3.89	96.31±1.45	96.59±0.91	97.67±1.06	87.84±2.07	93.32±3.30	96.35±2.14	96.49±1.82	94.28±2.05	96.99±1.58	**97.67±1.50**
11	76.01±31.77	70.63±15.75	87.86±9.98	98.12±1.71	98.77±1.02	79.49±10.21	90.97±4.93	93.26±4.45	91.04±6.20	97.77±1.15	96.74±3.52	**99.32±0.68**
12	92.88±14.78	98.82±0.86	99.75±0.32	99.87±0.24	99.89±0.24	98.40±1.04	99.25±0.48	99.90±0.16	**100.00±0.00**	99.99±0.02	**100.00±0.00**	99.90±0.13
13	89.09±27.74	**100.00±0.00**	99.81±0.30	97.04±4.71	98.24±2.97	94.79±5.77	97.85±3.36	99.28±0.55	99.15±0.69	96.77±3.98	99.13±0.84	99.84±0.49
14	97.59±3.48	99.20±0.44	97.24±2.02	98.07±1.39	98.84±0.87	94.89±3.16	96.61±3.34	99.08±0.51	96.07±4.10	95.42±3.18	99.05±0.48	**99.63±0.33**
15	71.21±12.29	74.37±5.31	86.87±4.40	87.63±3.47	92.60±2.39	75.41±4.48	84.51±2.90	88.89±5.61	88.85±4.51	89.54±4.17	90.70±3.59	**92.25±3.51**
16	85.88±11.62	89.12±5.01	96.73±1.98	97.17±2.11	**98.98±0.92**	91.18±4.19	95.07±3.43	97.14±2.81	97.78±1.20	98.05±0.65	96.47±1.73	98.57±0.51
OA(%)	86.81±6.25	90.91±0.74	94.57±0.76	95.88±0.37	97.11±0.47	89.65±0.76	93.50±0.63	96.02±0.56	95.68±0.69	95.54±1.05	96.09±0.88	**97.48±0.46**
AA(%)	86.08±10.40	92.86±1.26	96.23±0.94	97.58±0.37	98.11±0.33	91.60±1.08	95.29±0.50	97.33±0.47	96.82±0.64	97.06±0.74	97.29±0.59	**98.71±0.26**
$k \times 100$	85.27±7.02	89.87±0.84	93.95±0.85	95.42±0.42	96.78±0.52	88.48±0.85	92.76±0.70	95.57±0.63	95.19±0.77	95.04±1.16	95.65±0.98	**97.19±0.52**

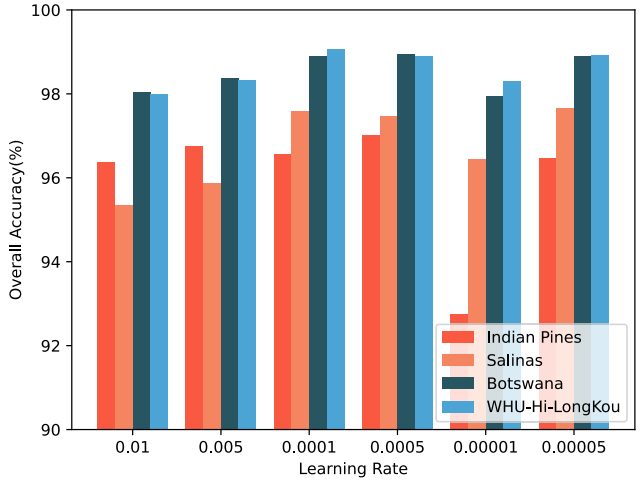


Fig. 13. Effect of six different learning rates on network performance on four datasets.

D. Classification Results and Analysis

In order to verify the effectiveness of the proposed method, nine representative network structures are selected for comparative analysis. These network structures include CNN-based 2DCNN and 3DCNN, and Transformer-based SSFTT, GAHT, MorphFormer, SpectralFormer, FactoFormer, MHCFormer, and SPTNet.

1) Analysis of Quantitative Results: In Tables V–VIII, we show the evaluation metrics including OA, AA, k, and single classification accuracy of the proposed method with all the compared methods on the four datasets: Indian Pines, Salinas, Botswana, and WHU-Hi-LongKou. The optimal results are highlighted in bold.

As presented in Table V, our method, utilizing only 5% of the training samples on the Indian Pines dataset, achieved an OA of 97.02% ± 0.47%. This represents an improvement of approximately 0.26% over the suboptimal MHCFormer. In comparison to CNN-based approaches, such as 2DCNN and 3DCNN, which yield reasonable results, these methods are limited in their ability to capture global information due to the restricted receptive field. Our method exhibits more consistent performance across nearly all categories when contrasted with Transformer-based methods. SpectralFormer, for instance, concentrates solely on the global modeling of spectral information during feature extraction, overlooking the importance of spatial information. Similarly, SSFTT, MHCFormer, SPTNet, and MASSFormer, while effectively extracting spectral–spatial features, do not adequately address multiscale features, which can impact classification accuracy. Our approach not only integrates the extraction of multiscale

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TABLE VII

PERFORMANCE COMPARISON OF TEN METHODS ON BOTSWANA DATASET

Class No.	CNNs		Transformers									Our Methods
	2DCNN [50]	3DCNN [36]	SSFTT [51]	GAHT [52]	MorphFormer [53]	SpectralFormer [54]	FactoFormer [45]	MHCFormer [46]	SPTNet [44]	DSNet [55]	MASSFormer [31]	Ours
1	99.96±0.11	99.89±0.24	99.89±0.24	99.96±0.11	99.30±1.08	**100.00±0.00**	99.19±1.27	99.85±0.25	**100.00±0.00**	**100.00±0.00**	96.22±3.18	99.70±0.40
2	84.75±8.95	90.30±6.00	97.62±3.66	98.22±4.72	99.01±1.47	98.32±2.73	98.71±1.66	86.93±18.22	99.70±0.63	94.85±6.43	96.93±1.68	**99.90±0.30**
3	89.12±2.78	98.21±1.31	99.68±0.43	**100.00±0.00**	99.04±1.63	91.99±3.81	94.30±4.54	99.48±1.43	99.36±1.46	99.12±1.56	97.09±2.88	99.92±0.24
4	85.35±8.81	97.26±1.03	99.44±0.93	99.81±0.37	**99.95±0.14**	70.93±12.02	98.93±1.44	99.77±0.56	99.86±0.42	99.07±0.98	98.84±1.97	99.86±0.30
5	86.88±6.26	83.75±5.51	96.17±3.99	96.10±4.94	92.83±7.21	25.87±5.46	90.22±5.76	94.20±5.72	**96.39±5.39**	93.09±2.57	89.37±5.13	95.28±3.43
6	81.34±8.53	75.20±8.81	93.38±7.18	**97.17±3.05**	95.17±4.28	74.57±10.62	91.90±5.16	95.58±6.91	97.06±2.30	94.80±5.85	90.63±4.21	96.80±2.32
7	95.98±2.58	99.54±0.51	99.69±0.38	**100.00±0.00**	**100.00±0.00**	97.95±1.73	98.34±1.58	**100.00±0.00**	**100.00±0.00**	99.92±0.15	98.15±2.39	99.92±0.15
8	78.62±8.86	87.00±8.25	97.73±3.14	99.90±0.30	98.97±2.62	76.21±7.84	92.46±10.67	96.80±8.38	99.46±1.47	99.90±0.20	93.89±7.58	**100.00±0.00**
9	78.89±6.76	95.92±2.49	97.42±2.32	98.54±3.98	98.76±1.52	66.53±5.79	95.19±2.03	99.14±1.02	98.95±2.16	98.60±1.22	99.43±0.84	**99.62±0.55**
10	96.25±2.43	96.69±2.09	98.31±1.61	99.40±1.04	99.80±0.41	91.77±5.21	98.06±1.98	99.56±0.87	99.84±0.27	99.19±0.36	98.79±1.84	**99.92±0.24**
11	96.95±2.24	97.84±2.16	97.77±3.28	98.33±2.41	96.07±5.96	97.97±1.47	98.07±1.57	99.31±1.04	**99.57±0.49**	96.26±2.32	96.56±3.18	98.00±2.33
12	96.24±2.28	99.28±0.86	**99.94±0.17**	99.39±0.46	98.73±1.66	95.30±1.45	98.67±1.51	99.06±2.64	99.61±0.70	99.34±1.33	95.25±3.75	99.06±1.02
13	97.61±1.00	98.81±1.60	99.03±1.63	**100.00±0.00**	**100.00±0.00**	96.79±2.62	99.18±0.96	**100.00±0.00**	**100.00±0.00**	99.40±0.73	99.78±0.45	**100.00±0.00**
14	88.63±5.86	86.21±5.08	93.47±5.49	97.79±3.28	98.32±5.05	83.47±0.67	88.74±6.72	89.26±7.67	95.05±4.82	96.63±4.28	95.89±5.23	**98.32±4.03**
OA(%)	90.07±0.94	93.73±1.47	97.96±0.79	98.92±0.71	98.15±1.22	82.44±0.76	96.03±1.40	97.93±1.08	**99.04±0.42**	97.96±0.52	96.23±1.15	98.96±0.32
AA(%)	89.76±1.17	93.28±1.62	97.83±1.02	98.90±0.79	98.28±1.03	83.41±0.80	95.85±1.28	97.07±1.57	98.92±0.49	97.87±0.48	96.20±1.03	**99.02±0.37**
$k \times 100$	89.24±1.02	93.20±1.60	97.79±0.86	98.83±0.76	98.00±1.33	80.98±0.83	95.70±1.52	97.76±1.17	**98.96±0.46**	97.79±0.56	95.91±1.25	98.88±0.34

TABLE VIII

PERFORMANCE COMPARISON OF TEN METHODS ON WHU-HI-LONGKOU DATASET

Class No.	CNNs		Transformers									Our Methods
	2DCNN [50]	3DCNN [36]	SSFTT [51]	GAHT [52]	MorphFormer [53]	SpectralFormer [54]	FactoFormer [45]	MHCFormer [46]	SPTNet [44]	DSNet [55]	MASSFormer [31]	Ours
1	98.08±1.76	98.35±0.64	99.60±0.23	99.72±0.25	99.64±0.30	99.37±0.50	99.59±0.20	99.63±0.34	99.40±0.36	99.75±0.15	99.66±0.29	**99.84±0.17**
2	80.64±13.31	82.65±7.88	96.32±3.72	95.15±2.21	97.41±1.66	81.06±7.15	92.05±3.30	98.64±0.71	95.17±1.86	88.90±10.46	95.09±3.49	**98.68±1.65**
3	0.01±0.04	56.89±7.31	86.63±6.66	85.25±5.65	91.96±3.59	83.71±6.67	89.59±3.75	91.73±3.66	89.77±5.43	85.54±7.80	94.41±3.28	**95.11±3.71**
4	94.73±2.09	97.39±0.90	98.51±0.77	98.79±0.46	98.93±0.50	96.65±0.66	98.81±0.40	**99.38±0.45**	99.07±0.35	98.11±0.29	98.69±0.52	99.17±0.36
5	2.33±5.15	29.35±14.00	84.46±8.90	85.86±10.14	89.18±7.07	58.56±15.51	78.53±6.64	88.13±6.11	83.10±6.05	84.17±8.96	83.03±13.04	**92.32±3.18**
6	63.69±31.62	97.41±2.76	98.09±1.99	98.63±1.74	97.41±2.41	93.60±3.78	98.52±1.43	98.86±0.95	99.19±0.52	97.14±2.58	**99.24±0.48**	99.03±0.84
7	99.73±0.62	99.94±0.08	99.92±0.10	99.86±0.13	99.88±0.16	99.89±0.07	99.65±0.36	99.93±0.06	99.92±0.16	**99.97±0.03**	99.72±0.17	99.91±0.12
8	44.39±29.58	88.91±3.99	91.98±3.61	92.46±3.66	89.51±5.41	75.13±6.77	81.77±4.26	93.48±2.93	87.57±6.73	88.52±2.30	89.74±5.16	**94.12±2.84**
9	44.47±19.51	69.36±9.01	89.52±5.43	89.68±5.79	82.98±6.79	44.18±5.75	85.40±4.08	**91.45±3.94**	82.56±3.44	87.92±2.64	88.65±6.15	90.81±3.62
OA(%)	88.24±2.98	94.79±0.49	98.12±0.30	98.22±0.35	98.18±0.42	94.30±0.52	97.44±0.41	98.79±0.18	97.97±0.35	97.50±0.80	98.13±0.50	**98.91±0.27**
AA(%)	58.68±6.91	80.03±1.97	93.89±1.20	93.93±1.85	94.10±1.01	81.35±2.02	91.54±1.42	95.69±0.88	92.86±1.67	92.22±2.79	94.25±1.71	**96.56±0.75**
$k \times 100$	84.36±4.01	93.12±0.65	97.53±0.39	97.66±0.46	97.60±0.55	92.49±0.69	96.63±0.54	98.41±0.24	97.32±0.46	96.71±1.06	97.54±0.65	**98.57±0.35**

spectral–spatial features but also considers the fusion of both local and global features.

In the Salinas dataset, as indicated in Table VI, our method attains the highest accuracy across several categories. While 3DCNN achieves a perfect accuracy of 100.00% ± 0.00% in the 13th category, likely due to the CNN's proficiency in local feature modeling, both MASSFormer and SPTNet also reach 100.00% ± 0.00% in the 12th category. However, their inability to effectively identify features at various scales results in lower OA, AA, and Kappa coefficients compared to our method. The 14th category of the Salinas dataset, which contains a limited number of samples, benefits significantly from our proposed MS2I2 module. This module captures multiscale spatial–spectral features effectively by integrating deep convolutions with convolution kernels of diverse sizes and orientations in the frequency domain, leading to the highest accuracy in this category—a 0.34% improvement over the next best method. Despite any potential shortcomings, our method, using merely 1% of the training samples, achieves an OA of 97.48% ± 0.46%, which is approximately 0.37% higher than the optimal MorphFormer.

On the Botswana dataset, as shown in Table VII, due to the dispersed distribution of features and the irregular shape of the categories, the CNN-based method is difficult to effectively model the global context information and is less sensitive to the recognition of objects at different scales. Therefore, our method achieves optimal single-class accuracy on multiple categories. Compared with popular methods such as SSFTT, SpectralFormer, and Morphformer, our method outperforms the OA by 1%, 16.52%, and 0.81%, respectively. Although our method is slightly lower than the optimal algorithm SPTNet by about 0.08% on OA and Kappa (k), it is higher by about 0.1% on AA. In terms of computation, our method is slightly inferior, but in terms of the number of parameters, it is about 12.314k less than SPTNet (the computation and parameters of other datasets will be analyzed in detail in the following). In addition, our method achieves an AA of 99.02% ± 0.37% using only 5% of the training samples. After ten repetitions of the experiment, the OA, AA, and Kappa of our method have the smallest fluctuation of 0.32%, 0.37%, and 0.34%, respectively, which further validates that the stability of our method outperforms that of other Transformer methods.

On the WHU-Hi-LongKou dataset, as shown in Table VIII, the highest accuracies were achieved by our method for almost all categories. It is observed that for the fifth category, narrow-leaved soybean (which is the same as the fourth category and has a high degree of similarity), our method achieves the best single-class precision despite the very small number of samples. This is due to the significant advantage of our method in recognizing feature classes at different scales. In addition, the DMC2R module in our method helps capture subtle spectral variations, thus distinguishing different features more effectively. In this dataset, we only used 0.2% of the training samples, and despite the limited sample size, the value of OA reaches 98.91% ± 0.27%, which is about 0.1% higher than the best method.

2) Visualization and Analysis of Results: In order to demonstrate more clearly the superiority of the proposed method over other comparative methods, Figs. 14–17 show the classification visualization results for all methods on the four datasets.

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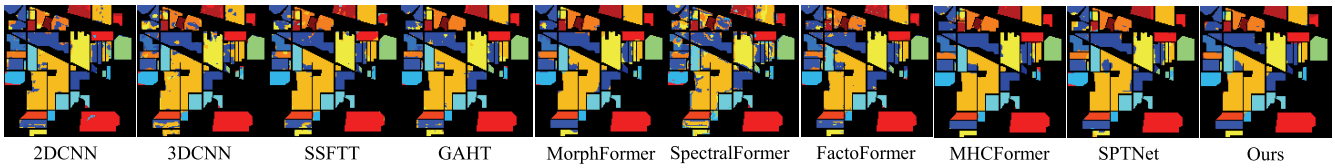


Fig. 14. Indian Pines dataset visualization classification map.

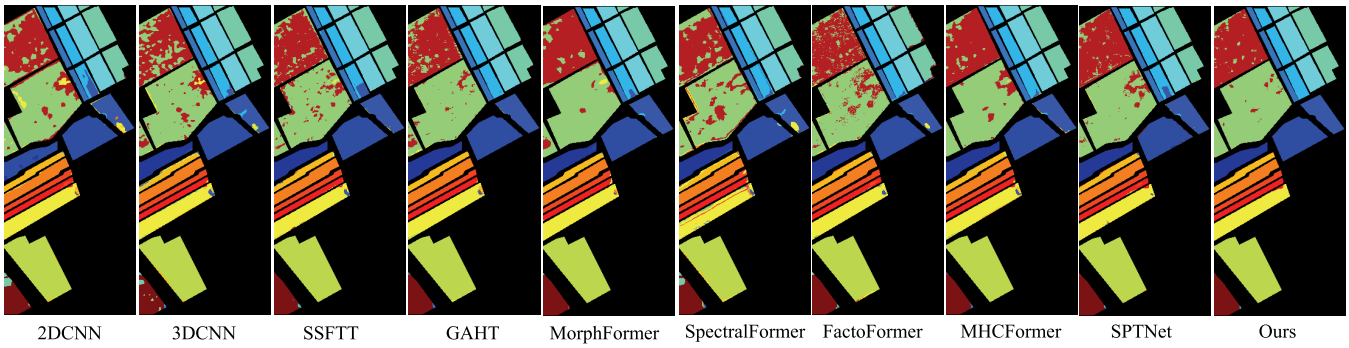


Fig. 15. Salinas dataset visualization classification map.

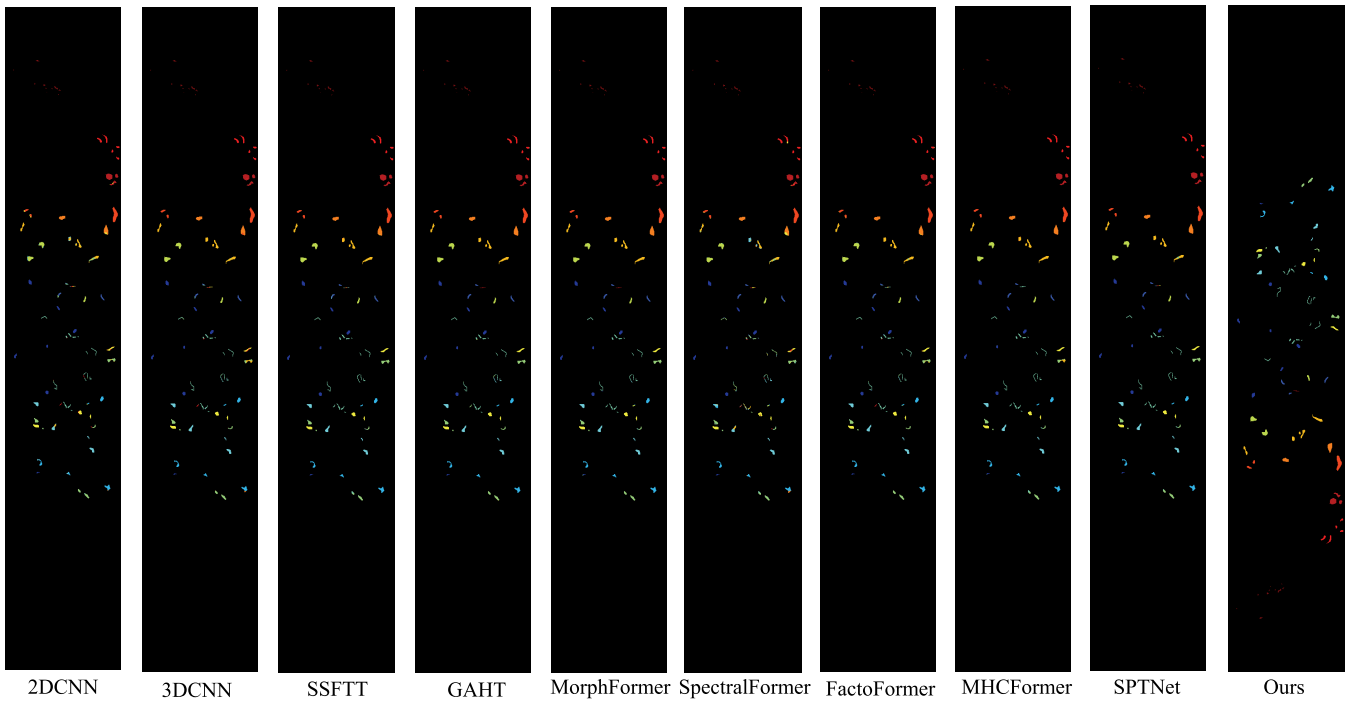


Fig. 16. Botswana dataset visualization classification map.

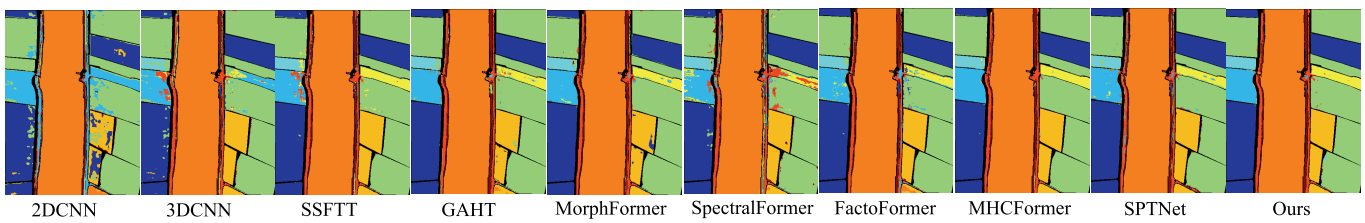


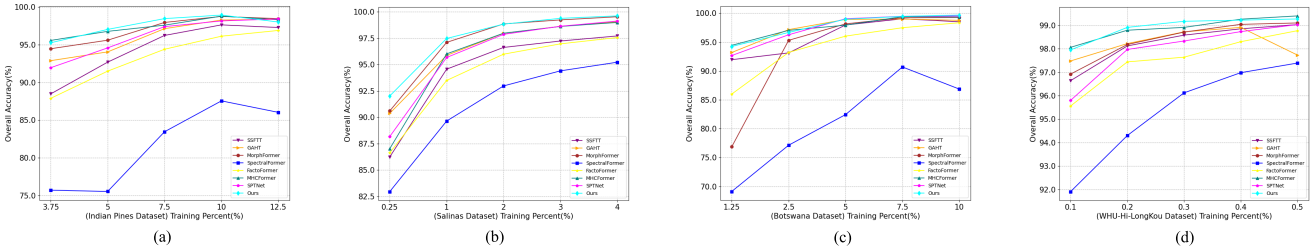
Fig. 17. WHU-Hi-LongKou dataset visualization classification map.

It can be clearly seen that the classification maps generated by our method MS2I2Former contain less noise and are closer to the real ground conditions. Taking the WHU-Hi-LongKou dataset as an example, it can be seen in Fig. 17 that for

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TABLE IX
COMPARISON OF THE NUMBER OF PARAMETERS AND COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS ON FOUR DATASETS

Methods	Indian Pines		Salinas		Botswana		WHU-Hi-LongKou	
	Parameters/K	FLOPs/MMac	Parameters/K	FLOPs/MMac	Parameters/K	FLOPs/MMac	Parameters/K	FLOPs/MMac
2DCNN	1708.944	66.441	1718.160	67.556	1581.966	51.108	1869.321	85.955
3DCNN	161.676	79.484	263.596	138.247	179.594	98.371	201.989	182.711
SSFTT	931.848	49.382	950.280	50.374	678.278	35.743	1253.953	66.738
GAHT	830.800	40.669	972.624	47.606	800.718	39.201	1514.121	74.148
MorphFormer	199.600	35.670	202.800	36.386	194.926	30.595	255.145	48.205
SpectralFormer	342.649	106.219	352.405	109.278	227.844	67.897	540.644	165.072
Factoformer	138.892	30.419	139.148	30.755	135.178	25.798	142.693	36.297
MHCFormer	3804.720	312.745	3892.608	318.236	2804.990	227.779	5108.665	419.425
SPTNet	173.616	**12.232**	175.664	12.422	145.326	**9.630**	209.001	15.543
Ours	**140.182**	12.620	**125.334**	**11.67**	**133.012**	12.17	**133.327**	**12.21**



(a) (b) (c) (d)

Fig. 18. Effect of different training sample proportions on OA. (a) Indian Pines dataset. (b) Salinas dataset. (c) Botswana dataset. (d) WHU-Hi-LongKou dataset.

the sixth golden-orange rice category, the classification maps obtained by our method are very clear on the surface and edges without noise. In contrast, the other nine comparison methods show varying degrees of noise on both the surface and edges of this category. The same is true for the other three datasets.

3) Calculation Cost Analysis: To highlight the advantages of our method MS2I2Former over the other compared algorithms in many aspects and its advantages are not limited to high accuracy, Table IX compares the number of parameters and FLOPs of our method with nine other methods. The results show that on Salinas and WHU-Hi-LongKou datasets, the number of parameters and FLOPs of our method are significantly lower than those of other models, indicating that we achieve high classification accuracy with a model with lower parameters and computational effort. On the Indian Pines and Botswana datasets, although the computational effort of our method is slightly higher than that of SPTNet, the FLOPs of our method are comparable to those of SPTNet, while the number of parameters of our method is significantly lower than that of SPTNet. Combining the analyses of the four datasets and the three dimensions of the number of parameters, the FLOPs, and the accuracy, the method we proposed has a clear advantage.

4) Robustness Analysis: In order to verify the robustness of our method, we conducted experiments on each of the four datasets with different percentages of training samples. Fig. 18 demonstrates the OA of each method with different training sample proportions. The experimental results show that our proposed method maintains a high classification accuracy with fewer training samples, while the accuracy of the other methods decreases. Taking the Indian Pines dataset as an example, even with a limited number of training samples, the classification accuracy of our method is higher than that of other methods with more training samples. This result further confirms the high robustness of our method.

TABLE X
ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODEL (BEST RESULTS IN BOLD)

Datasets	MS2I2 Block	FP Block	Evaluation Indicators		
			OA(%)	AA(%)	Kappa(k)
Indian Pines	×	×	96.32±0.70	92.38±2.32	95.80±0.80
	✓	×	96.33±0.36	93.06±2.19	95.82±0.41
	×	✓	96.99±0.66	93.83±2.16	96.56±0.75
	✓	✓	**97.02±0.47**	**94.28±1.86**	**96.60±0.54**
Salinas	×	×	96.31±1.10	97.62±0.82	95.89±1.23
	✓	×	96.86±0.42	98.38±0.27	96.51±0.46
	×	✓	97.26±0.41	98.48±0.21	96.95±0.45
	✓	✓	**97.48±0.46**	**98.71±0.26**	**97.19±0.52**
Botswana	×	×	97.72±1.17	97.69±1.02	97.53±1.27
	✓	×	98.53±0.69	98.43±0.81	98.40±0.75
	×	✓	98.93±0.72	98.99±0.73	98.84±0.78
	✓	✓	**98.96±0.32**	**99.02±0.37**	**98.88±0.34**
WHU-Hi-LongKou	×	×	98.20±0.42	94.59±1.89	97.63±0.55
	✓	×	98.40±0.31	94.89±1.39	97.90±0.41
	×	✓	98.77±0.34	95.44±1.26	98.38±0.45
	✓	✓	**98.91±0.27**	**96.56±0.75**	**98.57±0.35**

5) Ablation Analysis: In order to confirm the effectiveness of the two main blocks of our proposed method, we conducted ablation experiments on four datasets to evaluate the impact of each block on the classification performance. Our model consists of two main components: MS2I2 block and FP block. The evaluation metrics include OA, AA, and Kappa. The experimental results are shown in Table X.

Taking the Indian Pines dataset as an example, the analysis is as follows: the first row shows the results without any module; the second row shows the results with only the MS2I2 module; and the third row shows the results with the HSI 3-D stereo blocks directly input into the FP module. The results of the study show that the structure in the first row reaches the lowest value in all the metrics compared to the second and third rows, while the results in the second and third rows are significantly improved, which proves the significant effect

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of either of our designed modules. In addition, the results in the second and third rows show that the combined form of HC and MDCST in the FP module outperforms the MS2I2 module using only CNN. Combining the data in the second, third, and fourth rows, we find that the feature extraction ability relying only on a single module is lower than that under the combined effect of both modules, which emphasizes the importance of the MS2I2 and FP modules in extracting multiscale features. Ultimately, the data clearly shows that when both modules are present, the OA improves by 0.69% and 0.03% compared to the case where only the MS2I2 block or the FP block is added, respectively, and by 0.7% compared to the case where no module is added. This result again significantly demonstrates the effectiveness of the two main blocks in our method (the same is true for the other three datasets).

V. Conclusion

In this study, we propose the MS2I2Former model, which consists of an MS2I2 module, two FP blocks, a GAP layer, and an FC layer. The model integrates CNN and Transformer to deeply mine the multiscale spatial and spectral information in HSIs and reveals the correlation between spectral channels through covariance analysis, which provides a deep feature understanding for the model. In the MS2I2 module, we extract the multiscale features of HSI from different perspectives. In order to enhance the sensitivity of the model to the spectral information, we utilize the CM to explore the correlation between the spectral channels, which lays the foundation for the subsequent FP. The FP block mainly consists of HC block and MDCST. HC block includes GPWConv, GroupedConv, 3DCNN, and DWConv, which effectively enhances the extraction of local spectral spatial features in the band. MDCST combines MM and DCSA to realize the effective fusion of the local features with the global information. We conducted a large number of experiments on the four datasets, and the results show that the model still maintains a high classification accuracy with low parameters and low computation. In the future, we will continue to explore the advantages of CNN and Transformer to better apply multiscale representation to HSI classification tasks.

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